

## Declaration Sheet

**Award Title:** BSc(Hons) *Computer Science*

Declaration Sheet

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## **Abstract**

This report is written for a final year project I have been working on. A detailed analysis of a food delivery app named “Khanepa” will be provided in this report. This project was developed under the supervision of Mr. Anil Poudyal and our reader Mr. Sujan Shrestha. To begin with, the report discusses the introduction to the app, aims and objectives of the app, and then discusses the main functionalities and features of the app. "Khanepa" is especially made to meet the needs of people who are busy and offers them a time-saving solution. The app's user-friendly and intuitive interface makes it easy for users to understand and navigate through its features. Users can quickly comprehend the app's functionalities and make effective use of its services thanks to its simplicity. Restaurant owners have the option to add their businesses to the app, which will increase their visibility and draw in more customers. Customers can easily order food through the app, and it will be delivered to their chosen location. This report goes into great detail about the app's features and benefits for both users and restaurant owners. It covers a wide range of functions. In general, "Khanepa" aims to improve user experience, streamline the ordering process for food, and promote smooth interactions between diners and restaurants.

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# 1. Introduction

## 1.1. Project Brief

“The format of home delivery or the takeaways have gained plenty additional customers in locations like malls, offices and big-party orders for residential complexes.” (Gupta, 2019). Online food ordering and delivery refers to the process of placing an order for food through the internet and having it delivered to the consumer at the specified location. Customers can now conveniently order food through the internet and have it delivered to their preferred location, making online food ordering and delivery a rapidly growing and well-liked service. This trend has become very popular, especially with the younger generation and those who lead busy lives. In addition to being convenient for customers, online food ordering and delivery has benefited restaurants by making it easier for them to effectively handle and complete large and concurrent orders. In light of this, I have chosen to create the Khanepa food delivery app, realizing the enormous potential and beneficial effects this service can have on enhancing the general dining experience for customers while promoting the expansion and success of neighborhood restaurants.

Khanepa is designed to make it easier and more convenient for users to order food and find the best restaurants. It is always simpler to order food online or via a mobile device than it is to visit a restaurant, place an order, and then wait for it to be fulfilled. Online food ordering and delivery services have experienced a significant change because of the increased use of smartphones and the internet. People with busy schedules should use these services the most because they save a ton of time and effort. These kinds of services are the best for everyone in the modern world because most of the world has already gone digital and online. Similar online procedures are used by this app as well, enabling users to find restaurants and order food.

## **1.2. Aims**

Khanepa app's main goal is to provide users with quality service and make ordering food as simple as possible. With the ability to register and grow their business, the app aims to support small restaurants in response to the rising popularity of online food ordering. The app makes it simpler for users to discover new dining options by improving the convenience of finding nearby restaurants. The ultimate objective is to develop a user-friendly and effective application that is easy to use and extremely effective at satisfying user needs. The app aims to promote frequent usage and offer a seamless experience for all users by placing a priority on simplicity of use and intuitive design.



### **1.3. Objectives**

Khanepa app is made to make ordering food easier, especially for people with busy schedules or people who find it difficult to go to restaurants in person. The primary goal of the app is to increase restaurants' discoverability, making it simple for users to browse a variety of dining options. While the app has many objectives, its main goal is to make sure that customers have a super simple and easy experience when ordering food and having it delivered to their desired location. The app aims to give all users an easy and enjoyable food delivery experience by putting user convenience first and streamlining the ordering process.

## **1.4. Artefact**

### **1.4.1. Khanepa**

There are 3 different login types in this app which are mentioned as follows:

- Admin
- Restaurants
- Costumers

Only verified businesses can use this system because the admin is the only one who can create a restaurant account. On the other hand, people who want to look up restaurants and order food can easily set up customer accounts. The ability for restaurants to upload their menus is one of the platform's key features because it enables customers to browse the options and make appropriate selections. An order placed by a customer is sent right away to the appropriate restaurant for preparation. Customers can now follow the development of their order, ensuring that they are aware of the anticipated delivery date. Flexible payment options give customers the choice of paying online or when their order is delivered, depending on their preferences. The goal of this system is to make ordering food more convenient for both customers and restaurants. Customers can easily explore various dining options and place their orders with ease by centralizing restaurant menus and enabling online transactions. Restaurants profit from better visibility and more efficient operations. All things considered, the platform makes the process of ordering food easy and efficient for everyone involved. For online payments, API of digital wallets will be used. Customers can also rate the restaurants for their foods and services and can also leave feedbacks/comments.

Some of the key points to mention:

- This app also allows customers to rate and give feedbacks to the app to help improve the experience.
- The app will display all the registered restaurants which allows customers to choose from various options.
- A very simple mobile application which even people with minimal knowledge can use easily.

### **1.5. Problem Statement**

People with busy schedules frequently encounter difficulties when it comes to personally visiting a restaurant, placing their desired food order, and then patiently waiting for it to be prepared in today's fast-paced and increasingly digitalized world. Time-consuming, inconvenient, and not always practical, this process can be. The creation of a mobile application aims to offer a workable solution to these problems. People are constantly looking for more practical alternatives for a variety of tasks, including ordering food, as a result of the explosive growth in the use of the internet and mobile devices. The conventional method of going to a restaurant and waiting for the order to be finished is unappealing to many people. People might not be aware of all the fantastic restaurants in their area or the wide variety of dishes they offer on their menus. Their ability to experiment with new culinary experiences is restricted by this ignorance. Furthermore, people frequently have little free time due to busy schedules, making it difficult to pick up their orders in person. The mobile application offers a complete solution in this situation. The app offers a user-friendly platform that enables customers to quickly browse through various restaurants, access their menus, discover new dishes, and conveniently place their orders by leveraging the power of technology and connectivity. Additionally, the app makes it easier for customers to get their orders delivered to the place of their choice, saving them both time and effort. These issues are more prevalent than ever in the modern world, where time is valuable, and convenience is highly prized. Because of this, the mobile application aims to make ordering food for busy people simpler, eliminate related difficulties, and improve their overall dining experience.

## **1.6. The project as a solution**

Khanepa has been created as a practical response to the aforementioned problems. Users of the app can conveniently order food online and discover the best restaurants in their area. Users can avoid physically going to the restaurants and save a lot of time and effort as a result. Customers have a variety of options to choose from thanks to the app's display of a variety of registered restaurants. Users can access the restaurant's food menu by clicking on any of the listed restaurants, giving them a thorough view of the available dishes. This gives users the power to choose wisely based on their preferences. Customers can use the app to place their order after choosing the desired items. By providing these features, the app seeks to simplify the entire food ordering process, improve user experience, and offer users a simple and effective platform to indulge their cravings for delicious food.

## **1.7. Scope and Limitations of the project**

Every program has its own strength and flaws. Likewise, this app is also designed to work with certain conditions only. Some of its Scope and Limitations are mentioned below:

### **1.7.1. Scope**

- The app is very easy to use and is understandable even with slight ideas.
- To find nearby restaurants for users to choose from
- Deliver foods as per orders
- Save time and energy for people

### **1.7.2. limitations**

- Cannot show exactly how much time is required for the delivery.
- Is not able to show the best restaurants for every individual user differently.

## **1.9 Report Structure**

This report contains of the following topics:

### **1.9.1 Introduction**

This section provides a broad overview of the mobile application, the problem domain, and the project topic as a proposed solution to the identified problem. This section discusses the project's aims and objectives, as well as the artifacts to be developed, potential scopes, and limitations.

### **1.9.2 Literature Review**

This section contains the research and investigation conducted for the project, including the literature review that was conducted. It provides a comparative and evaluative analysis of similar mobile applications to gain insights and understand their strengths and weaknesses.

### **1.9.3 Main Body**

This section includes the project's design documents, like sketches and diagrams, that outline how the application will look and function. It also explains the tools and methods used for development and provides detailed information about the developed features. The section also describes the approach followed during development, including any testing conducted under various conditions if necessary.

### **1.9.4 Conclusion and Evaluation**

In this section, the report and the developed project are summarized and evaluated. It includes a conclusion and highlights the key findings and outcomes of the project's development process.

## **2. Literature Review**

### **2.1 Some Related Works**

#### **2.1.1. Foodmandu**

Recently, many of the smaller and bigger projects related to online food ordering and delivery services have been highly focused. Even in Nepal, the trend of online food ordering and delivery services are getting a very good response. Among many, “Foodmandu” is one of the most rapidly growing food delivery services in Nepal right now. It is known as the first company in Nepal to provide food delivery service. “Manohar Adhikari founded the company in 2010. He got an idea of a food delivery business based on his hardship finding decent eateries nearby when he was working for a software company in Kathmandu.” (Shah, 2020). The company is currently serving cities like Kathmandu, Pokhara, Lalitpur, and Bhaktapur and plans to expand its services all over Nepal soon. Although food delivery platforms like Bhoj, Pathao Foods, and many others are competing strong, Foodmandu is still leading due to its trust, experience, stability, and quality. (Shah, 2020). “Using its website and social media platforms, Foodmandu delivers foods upon order from popular restaurants in Kathmandu at consumers’ doorstep.” (Rai, 2022). Like any other Food Deliver App, Foodmandu also lets users choose from different restaurants and place their order. Users can also search and choose food items directly and find restaurants serving those items. “Companies like Foodmandu work with two active sets of actors: the delivery riders and the actual restaurants.” (Rajbhandari, 2022). Foodmandu partners with restaurants so that the restaurants get registered to the app and start their business. Restaurants pay the food delivery app a small percentage of every order they sell through the app. (Rajbhandari, 2022). Delivery is done by the delivery drivers. Currently, Foodmandu employs both full time as well as part time delivery



riders. “Foodmandu currently employs over 190 delivery riders and has partnered with over 800 restaurants across the capital and Pokhara” (Rajbhandari, 2022). This has somehow helped for the employment of some people.

Impact and benefits of Foodmandu's services in Nepal:

- **Time and Convenience Savings:** Customers have benefited greatly from the rise of online food ordering and delivery services like Foodmandu because it gives them a convenient way to order food from their preferred restaurants. Customers can browse through a variety of restaurants and food options on their mobile devices with just a few taps, place their orders, and have their meals delivered right to their door. By doing this, they are able to save both the time and effort they would have otherwise spent cooking at home or traveling to a restaurant.
- **Increased Restaurant Visibility and Reach:** Foodmandu's platform gives restaurants the chance to grow their clientele and increase their level of exposure. Restaurants can access a sizable user base and take advantage of Foodmandu's marketing and promotional initiatives by partnering with the platform. By increasing their visibility, restaurants are able to draw in new customers who may not have previously known about their offerings, ultimately increasing their sales and revenue.
- **Economic Impact and Job Creation:** The rise of food delivery services like Foodmandu has improved employment conditions in Nepal. Foodmandu has helped many people find employment and opportunities for income thanks to its employment of over 190 delivery drivers and partnerships with over 800 restaurants. Both full-time and part-time delivery drivers have found work, and thanks to their relationships with restaurants, they have been able to stay in business and make money even during difficult times.
- **Technological Developments:** The success of Foodmandu also demonstrates how far the food industry in Nepal has come in terms of technology. Customers, restaurants,

and delivery drivers can interact and transact on the company's website and mobile app in an efficient manner. From ordering to delivery, the use of technology reduces the entire process, ensuring accuracy and efficiency.

- **Positive Customer Feedback and Satisfaction:** Foodmandu has gained the loyalty and trust of its customers thanks to its emphasis on quality, stability, and experience. The glowing testimonials and constructive criticism from pleased clients also help to build the company's reputation and success. Foodmandu has won the favor of many Nepalese by offering dependable delivery, a wide variety of restaurant options, and upholding high standards for food quality.

Overall, customers, restaurants, and the economy have benefited greatly from Foodmandu's entry and expansion into Nepal's online food delivery market. Its position as the top food delivery service in the nation has been cemented by its creative strategy, extensive network of restaurant partnerships, and dedication to customer satisfaction.

### **2.1.2. Bhojdeals**

Bhojdeals is another one of the best food delivery companies in Nepal. It is popularly known and “Bhoj”. One of its best features, just like any other food delivery apps, it traces the location of the user and shows the available restaurants within the traced area. It is affiliated with over 400 restaurants and has expanded its reach to Kathmandu, Bhaktapur, Lalitpur, Butwal, and Bhairahwa (theeverestlist, 2022). Bhojdeals, like many other food delivery apps, saw a significant increase in popularity after the COVID-19 lockdown. Bhojdeals follows a similar working schedule to Foodmandu. Through the app, users can conveniently choose from a range of restaurants in their neighborhood and order the meals they want. The app sends users timely notifications after an order is placed to keep them updated on its status. Users are given a sense of transparency and assurance thanks to this tracking feature, which makes sure they are informed of the progress of their orders. The Bhoj Wallet feature on

Bhojdeals also provides its users with an extra perk. Users receive credits in their Bhoj Wallet for each successful order, which they can then use to pay bills or conduct transactions inside the app. This not only improves the user experience overall but also gives users a reason to keep using the app for their food ordering requirements. Bhojdeals strives to offer users a seamless and rewarding experience, making it a popular option among food delivery apps thanks to its user-friendly interface, reliable order tracking, and the bonus of the Bhoj wallet.

Advantages and unique features of BhojDeals:

- A wide variety of restaurant options are available to users thanks to Bhojdeals' partnerships with more than 400 eateries. Bhojdeals makes sure that customers have a variety of options to suit their preferences and cravings, from neighborhood eateries to well-known restaurants. Bhojdeals stands out thanks to its extensive network of restaurant partnerships, which also increase its appeal to a wide range of customers.
- Beyond Kathmandu, Bhojdeals now offers services in places like Bhaktapur, Lalitpur, Butwal, and Bhairahwa. With this extension, more people across Nepal will be able to take advantage of the ease of online food ordering and delivery through the Bhojdeals platform. Bhojdeals serves a wider customer base and helps the online food delivery sector in these cities grow by expanding its reach.
- Bhojdeals offers users a customized and effective experience by utilizing location-tracking technology. The app uses the user's location to determine which restaurants are nearby, making it simpler for users to find nearby dining options and conveniently place orders. By saving time and providing a more localized selection of restaurants, this feature improves the user experience.
- Through prompt notifications, Bhojdeals keeps users updated on the status of their orders. Transparency is ensured by the order tracking feature, which enables users to follow the status of their orders from the time they are placed until they are delivered. By being transparent, Bhojdeals is able to gain users' trust and confidence while also giving them peace of mind.

- Bhojdeals distinguishes itself from competitors by providing users with the Bhoj Wallet feature. Users are encouraged to keep using the app for their food orders by earning credits in their Bhoj Wallet for each successful order. The ability to use these credits for in-app purchases or bill payments enhances the overall user experience and promotes customer loyalty for Bhojdeals.

Due to its extensive restaurant selection, expansion into multiple cities, use of location tracking, open order tracking, and the extra benefit of Bhoj Wallet rewards, Bhojdeals stands out among Nepal's food delivery apps. Because of these features, using Bhojdeals to order and have food delivered online in Nepal is an easy and satisfying experience.

## **2.2 Technicality of the project**

### **2.2.1. Flutter**

“Flutter is a simple and high-performance framework based on Dart language, provides high performance by rendering the UI directly in the operating system’s canvas rather than through native framework.” I decided to use Flutter as I wanted to try and learn something new and also challenge myself.

### **2.2.2. Firebase**

Firebase is a Backend as a Service platform for mobile and online app development that offers developers tools and services to aid in their development efforts. Firebase is a real-time database that is stored in the cloud and allows users to sync data in real-time while also storing data for future use. Apps connected to Firebase are connected over WebSocket rather than traditional HTTP. Real-time synchronization allows users to access data from any location and makes user collaboration simple and quick. Files in Firebase are stored in binary format. It ensures network resilience and security.

### **2.2.3. Online Food Delivery Platforms and their Impacts on Sustainability**

The benefits of online food delivery (FD) were evident during the worldwide 2020 COVID-19 epidemic, as it allowed consumers to maintain access to prepared meals and allowed food providers to remain in business. There are others who are opposed to online FD, and there have even been allegations of consumer and restaurant boycotts because of it. So, it's time to pause and think about how internet FD is affecting the world at large and who it's affecting. This article also talks about the most recent research in this sector, indicating a slew of beneficial and bad effects, and frames the analysis via the three pillars of sustainability. Financially, online FD offers position and business growth potential, but it has also been attacked with the exorbitant commission it charges restaurants and the dubious working conditions for delivery personnel. When seen from a sociological standpoint, FD on the internet modifies customers' relationships with food and has an impact on public health plus transportation networks. The garbage produced is substantial, and so are the carbon footprints it leaves behind. As time goes on, stakeholders in online FD will need to think about how to both limit the bad effects and maximize the favorable ones (Li, et al., 2020).

### **2.2.4. Benefits of Food Delivery App**

Below mentioned points are some reviewed notes from an article titled “A Study on Impact of Online Food delivery app on Restaurant Business” published by Dr. Mitali Gupta in 2019.

### **2.2.5. Easy to Use and Understand**

Due to their ease of use and the time and effort they save for users, mobile apps for ordering meals are becoming more and more popular. More and more people are using their smartphones to order food for delivery due to the convenience of these apps. These apps are the ideal choice for ordering meals while on the go due to their functionality, usability, intuitive interfaces, smooth navigation, and robust search features. On their mobile devices, users can quickly browse menus, choose their preferred items, tailor their orders, and complete the transaction. Customers benefit from a seamless and effective process that allows them to save time and effort while still receiving what they need. (Gupta, 2019).

### **2.2.6. Flexible Payments**

Mobile food ordering apps work to improve the user experience by providing a variety of payment options to suit different preferences. These apps incorporate numerous significant payment channels into their platforms because they recognize the value of frictionless and secure transactions. These apps enable users to select the payment method that best suits their convenience and financial preferences by supporting a variety of cashless options like credit cards, debit cards, mobile wallets, and online banking. This flexibility not only makes ordering simpler, but it also encourages a cashless society and raises client satisfaction. These meal ordering apps make sure that customers have a variety of payment options at their disposal, improving the overall ordering experience and making it more seamless and enjoyable, regardless of whether users prefer the simplicity of digital wallets, the security of card payments, or the convenience of online banking (Gupta, 2019). Many food delivery applications also provide Cash on Delivery services.

### **2.2.7. Order Tracking**

Thanks to the integration of GPS navigation devices, both the customer and the delivery person can have precise or approximate knowledge of when the meal will be delivered. This technology enables real-time tracking of the delivery person's location, allowing the customer to stay informed about the progress of their order. As a result, the customer can have a better estimation of when their meal will arrive, providing them with convenience and transparency in the delivery process. (Gupta, 2019).



### **2.2.8. Effective Customer Service**

Mobile apps with 24/7 customer service offer great assistance to consumers by immediately responding to their enquiries, problems, and complaints. This improved customer care system ensures that customers and support executives communicate efficiently and effectively. Users may easily interact with customer service specialists who are accessible to provide guidance and address any issues with just a few clicks on their smartphones. This simplified procedure allows customers to obtain prompt and personalized assistance, which improves their overall experience with the mobile app. (Gupta, 2019).

### **2.2.9. Comparison**

Comparing my app with the points mentioned above, I can say that

- My app is easy to use and understand by people with minimum knowledge as well.
- Payment Methods include online payments like eSewa and Khalti. Cash on Delivery will also be an option.
- One can also place orders efficiently.
- In case of any difficulties, Customer Service will also be available. A complaint can be submitted via feedback.

### **2.2.10. GPS**

Geolocation is the process of determining and monitoring the location of connected electronic devices through the utilization of position technologies such as IP addresses and GPS. Mobile applications with geolocation capabilities create a smooth link between the device and the satellite, allowing for the collection of data in real time. The data are useful for determining the behavior of a moving object or person, and businesses may put them to use in the process of developing new strategies (Technologies, 2018). A number of different technologies, including WiFi, Geofencing, Cell ID, and GPS, are utilized in the process of information collection. The most convenient aspect of mobile applications that are based on geolocation is that they direct users to their destination by the quickest route (Technologies, 2018).

### **2.2.11. Google Maps API**

Google Maps is a web-based mapping service created by Google which gives precise information on a geographical area. It offers satellite services, aerial photography, street maps, and 360-degree street (TechTarget, 2019). The Google Maps API is a flexible tool for developers, providing a number of APIs to satisfy various needs. The Google Maps API opens up countless opportunities for improving user experiences and integrating useful location-based functionality into websites and mobile applications, from calculating distances and displaying detailed place information to offering virtual tours and turn-by-turn directions. Developers can build engaging and user-friendly platforms with robust mapping and navigation features by utilizing the power of Google Maps, facilitating users' exploration and engagement with their surroundings. (TechTarget,2019).

People can also view maps and locate the restaurants of their choice which are registered on the application. Also nearby restaurants can be searched.

### **2.2.12. Payment Method**

The ability to pay for orders placed through food delivery apps often involves the usage of several different payment methods. The most typical way to make a payment is by using a credit or debit card, which must first be input into the app before it can be saved for later use (Ghosh, 2021). Numerous apps for food delivery also provide alternative payment methods in addition to credit or debit cards. These could include digital wallets that let users use their smartphones or other compatible devices to make payments, like Apple Pay or Google Pay. Other electronic payment methods, like online banking or payment gateways like PayPal, may also be supported by some apps. These additional options give users more options and consider different preferences or situations. Food delivery apps aim to accommodate a wide range of user preferences and ensure a quick and secure transaction process by providing a variety of payment options. These apps aim to give users a simple and convenient way to pay for their orders, enhancing the overall user experience, whether it be through conventional card payments or newly emerging digital payment solutions. A user may be able to link their app account to a rewards program or loyalty card through some applications. This provides the user with the opportunity to accumulate points or receive discounts on their purchases (Ghosh, 2021).

When creating their app's payment system, the developers consider the most widely accepted and favored payment options. Credit cards and debit cards typically fall under this category because of their widespread acceptance and universal recognition. Users can conveniently make payments with just a few taps by securely linking these cards to the app. Most applications provide users with a variety of payment choices to cater to their unique interests and requirements (Shaliza, 2019).

For my app, I've decided to use API of online wallets like eSewa and Khalti for online payments while "Cash on Delivery" will also be an option.

### 3. Project Methodology

“The Agile methodology is a way to manage a project by breaking it up into several phases.” (wrike, 2022). Compared to the conventional waterfall method that was common before the 2000s, the agile methodology has proven to be significantly more efficient and advantageous. Using a sequential and linear process, the waterfall method required that each stage of development be finished before moving on to the next. This method's major flaw was that user feedback couldn't be collected until the finished product was delivered. Because of the distance this caused between the development team and the end users, it was challenging to incorporate changes or handle problems as they arose. In contrast, the Agile methodology introduced iterative and incremental practices, which completely revolutionized the software development sector. Customers and stakeholders actively participate in the development process with Agile, offering ongoing feedback and insights. Greater collaboration, openness, and flexibility are made possible by this. Customer satisfaction is prioritized by agile methodologies like Scrum and Kanban by producing functional software in brief sprints. This iterative approach makes frequent releases possible and eliminates the need for extensive reworks later in the development cycle by letting users see the progress, offer feedback, and suggest changes up front. Agile also places a strong emphasis on continuous improvement, which empowers teams to adjust to changing requirements, produce high-quality products, and shorten time-to-market. Agile methodology, in general, ensures closer customer and developer collaboration, which improves product quality, customer satisfaction, and delivery speed. Agile methodology fulfils all those requirements (wrike, 2022). Among all the widely used agile methodologies, I choose Scrum.

“Scrum is a framework for developing and maintaining product development using agile methodology and based on below characteristics, namely, Built-in instability, Selforganization, Overlapping development phases, Multi-learning, Subtle control,

Organizational transfer of learning” (wrike, 2022). I used Scrum as my program will undergo iterative product development. Here, I will show you how iterative development works on one of the features of the application.

**Rating Restaurants and their Foods:**

Iteration 1: Listing the restaurants

Iteration 2: Being able to search the restaurants

Iteration 3: Search Filters

Iteration 4: Giving feedbacks and ratings

### 3.1. Scrum

Scrum is a widely used framework for managing complex projects that offers a structured and flexible approach. It falls under the category of agile methodologies, which place a premium on adaptability, teamwork, and continuous development.

A small, cross-functional team that collaborates closely to accomplish a common objective forms the basis of Scrum. This team works closely together over a set period of time called a sprint, which is typically a few weeks long. To facilitate communication and progress monitoring, regular meetings are held. Daily stand-up meetings give team members the opportunity to communicate progress on their projects, talk about any difficulties they may be having, and make sure everyone is on the same page. To specify the precise tasks and objectives to be completed, sprint planning meetings are held at the start of each sprint. At the conclusion of each sprint, the team gathers for a retrospective meeting to discuss its progress, pinpoint areas for improvement, and make any necessary modifications to improve subsequent sprints.

Different artifacts are used by Scrum to support the development process. A dynamic repository of requirements, the product backlog is a prioritized list of all the work that needs to be done. The specific tasks to be accomplished during the current sprint are listed in the sprint backlog, which is derived from the product backlog. The team creates a potentially shippable increment of the product at the conclusion of each sprint, ensuring that valuable deliverables are consistently produced throughout the development lifecycle. Scrum is especially well-liked by software development teams because it can accept changing requirements and provide incremental value. Teams using Scrum can react quickly to changing customer and market demands because it places a strong emphasis on collaboration, transparency, and adaptability. It promotes a culture of continuous improvement, empowering teams to gain knowledge from their mistakes and improve their procedures in order to increase productivity and satisfy customers.

## **4. Tools and Technologies**

### **4.1 Flutter**

“Flutter is a simple and high-performance framework based on Dart language, provides high performance by rendering the UI directly in the operating system’s canvas rather than through native framework.” (tutorialspoint, 2022). Flutter is a modern and effective framework that gives programmers the tools they need to make reliable and aesthetically pleasing applications. Flutter, which is based on the Dart programming language, provides a simple and efficient development environment. One of its main advantages is that it can achieve high performance without the use of intermediary native frameworks by rendering the user interface (UI) directly on the canvas of the operating system. Flutter gives developers a wide range of customizable widgets and a large library of pre-built components, enabling them to design UIs that are consistent and aesthetically pleasing across various platforms. The reactive programming paradigm-based architecture of the framework enables simple UI updates and effective state management. Flutter’s “write once, run anywhere” logic enables programmers to create applications for a variety of platforms, such as iOS, Android, the web, and desktop, using a single codebase. As a result, maintenance expenses, development time, and effort are all decreased. I decided to use Flutter as I wanted to try and learn something new and challenge myself.

### **4.2. Firebase**

“Google Firebase is Google-backed application development software which allows developers to develop Android, IOS, and Web apps.” (javatpoint, 2022). Google Firebase is comprehensive and user-friendly. For the development of applications for the Android, iOS, and web platforms, it offers a set of tools and services. The functionality of applications is



improved and the development process is made simpler by a number of the features that Firebase provides. The real-time database provided by Firebase, which enables programmers to create applications with real-time data synchronization, is one of the service's main benefits. This enables real-time collaboration and effortless usage because changes made to the database are immediately reflected across all connected devices. Firebase provides authentication services that make it simple for developers to integrate secure user authentication and authorization into their applications. This eliminates the requirement to design intricate authentication systems from the ground up and offers a safe and dependable solution. Firebase can power the backend of an app, including data storage, user authentication, static hosting, and other features (Singh, 2018). Firebase has analytics and crash reporting tools that offer useful insights into how users use applications, their behavior, and performance problems. Making informed decisions, enhancing user experiences, and enhancing application stability can all be done with the help of this data.

## 5. Artefact Designs

### 5.1. SRS Legend

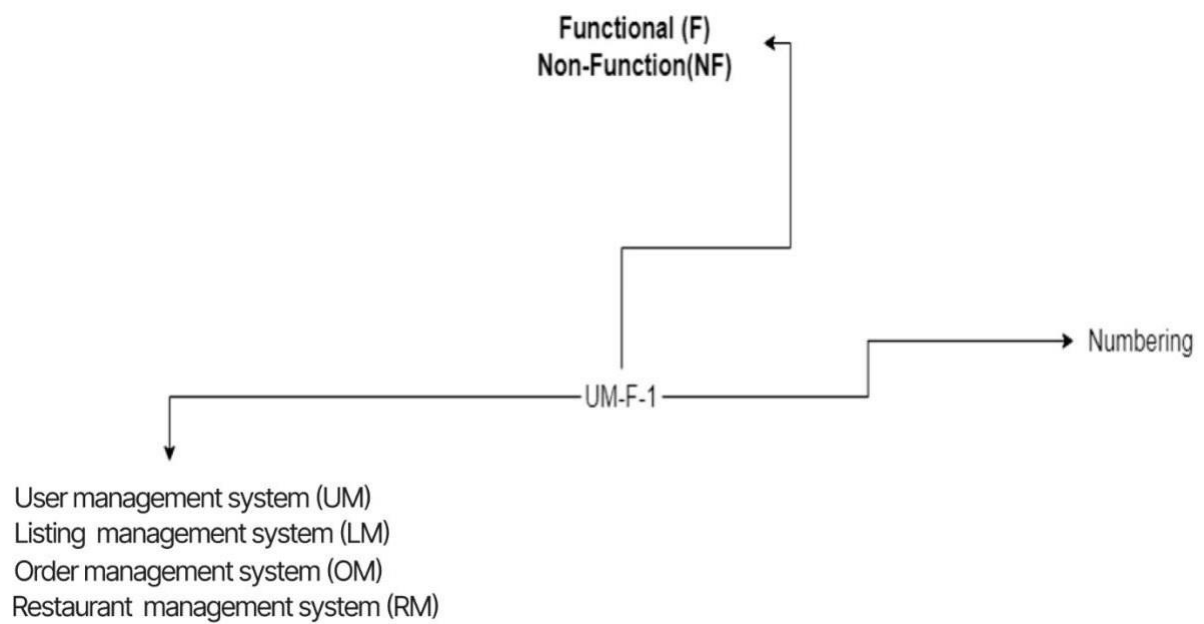


Figure 1 SRS Legend Figure

## 5.2. User Management System (UM)

### SRS

| Req Code. | Req. Desc                                                                                                                             | Testing ID |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------|------------|
| UM-F-1    | The system should allow the customer to self-registration and login. It should support social registration too (Google, Phone number) | TC-01      |
| UM-NF-1.1 | The password for the registration must be a combination of alphabets, numbers, and special characters.                                | TC-02      |
| UM-F-2    | The admin should be allowed to approve the restaurant's registration.                                                                 | TC-03      |
| UM-F-3    | User should have their profile with their all information.                                                                            | TC-04      |
| UM-F-4    | Users should be able to view nearby restaurants.                                                                                      | TC-05      |

|        |                                                                                 |       |
|--------|---------------------------------------------------------------------------------|-------|
| UM-F-5 | Restaurants should be able to add foods to their menu.                          | TC-06 |
| UM-F-6 | Customers should be able to order foods from the restaurants.                   | TC-07 |
| UM-F-7 | Customers should be able to leave a feedback for the service of the restaurant. | TC-08 |

## Activity Diagram

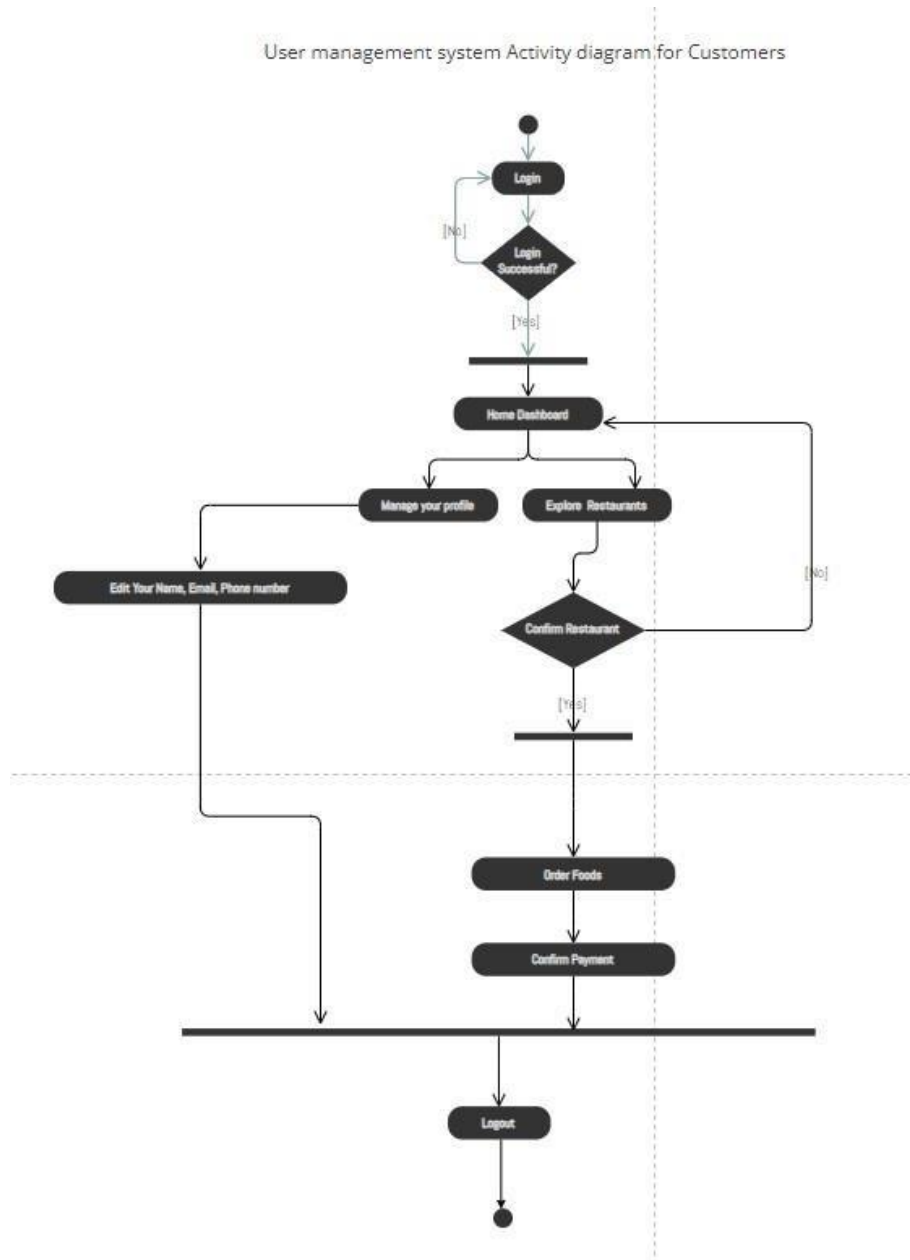


Figure 2 Activity Diagram for Customers UM

User management system Activity diagram for Restaurants

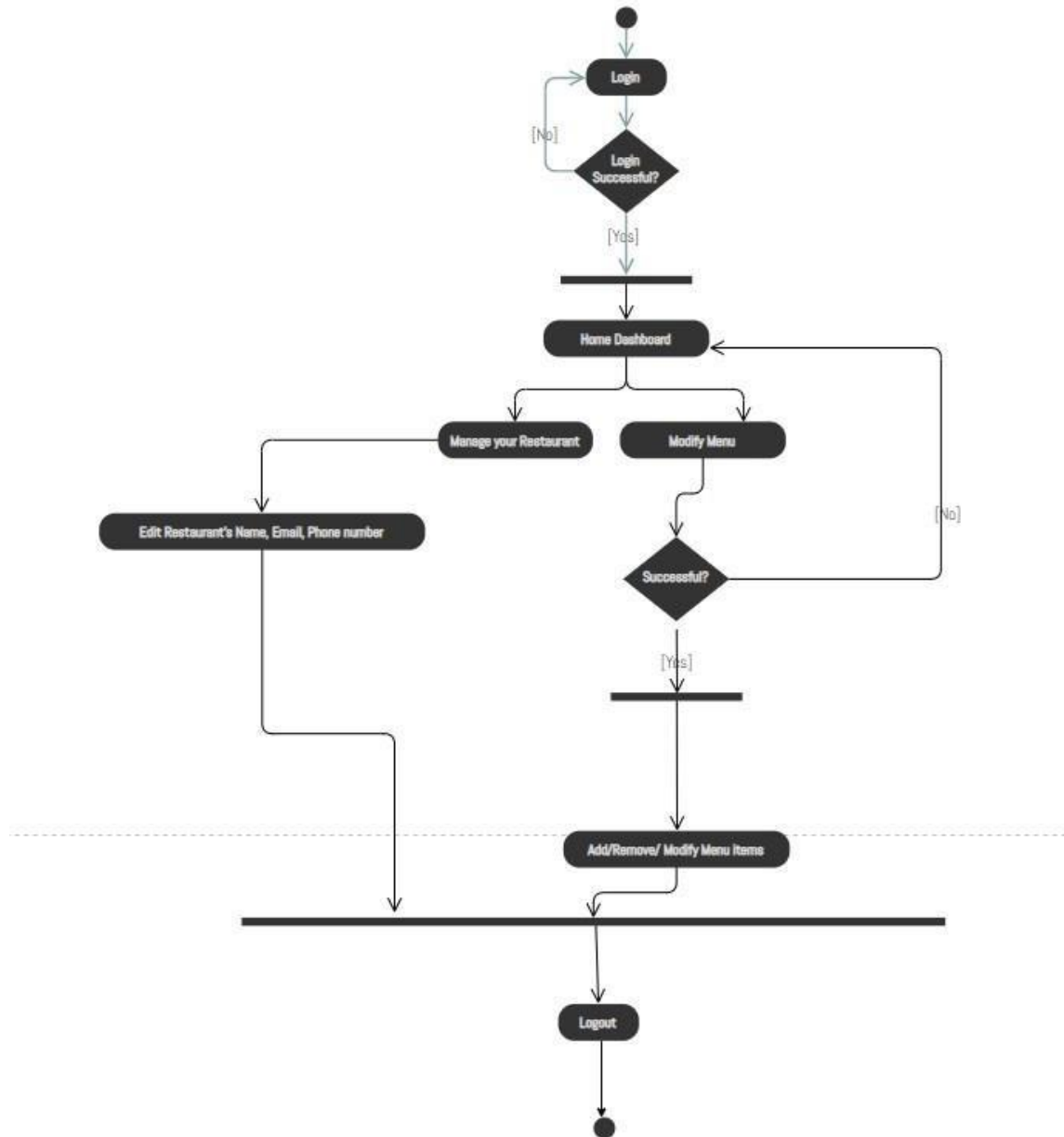


Figure 3 Activity Diagram for Restaurants UM

## Use Case Diagram

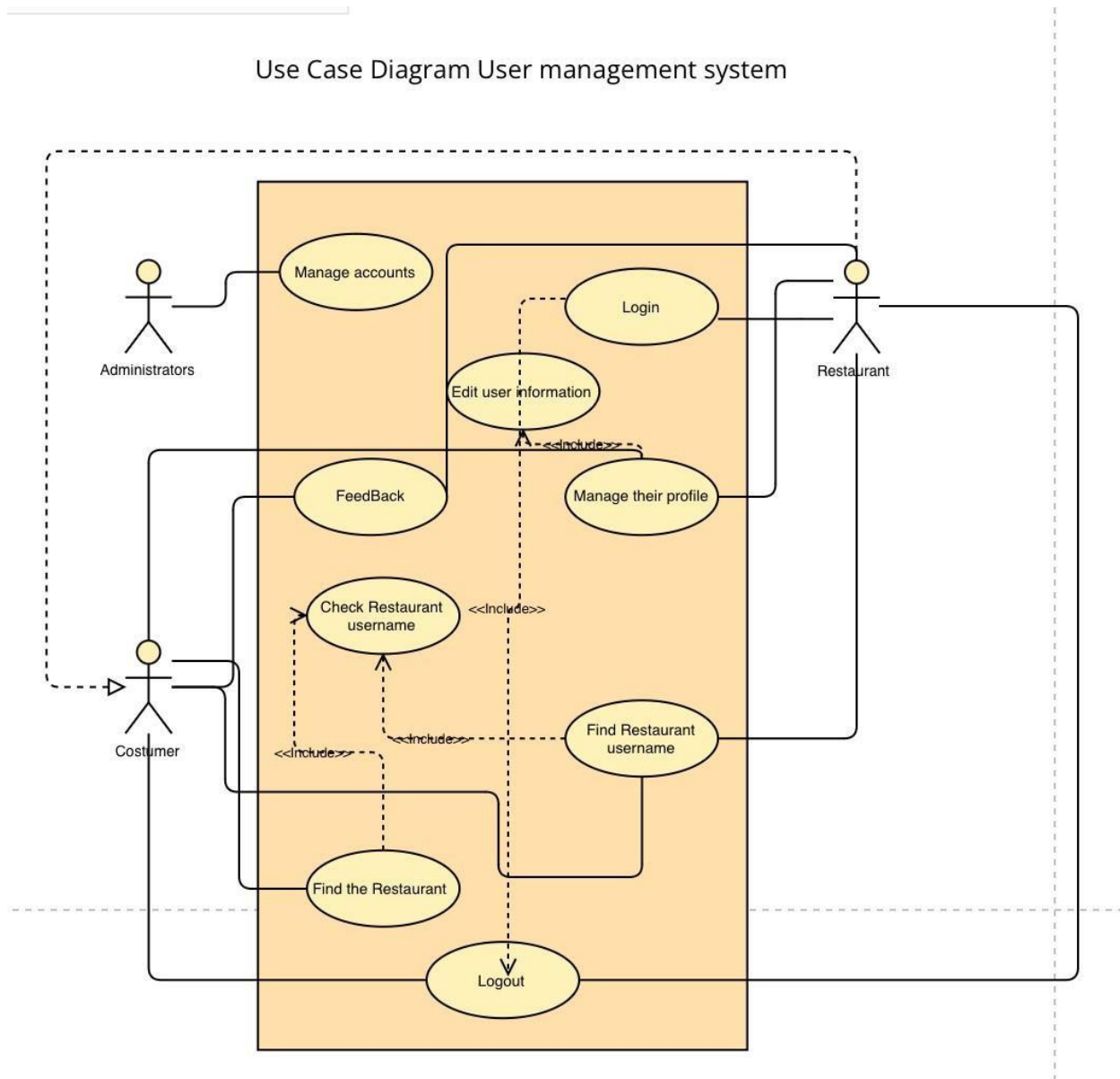


Figure 4 Use Case Diagram User management system

## Entity Relation Diagram (ERD)

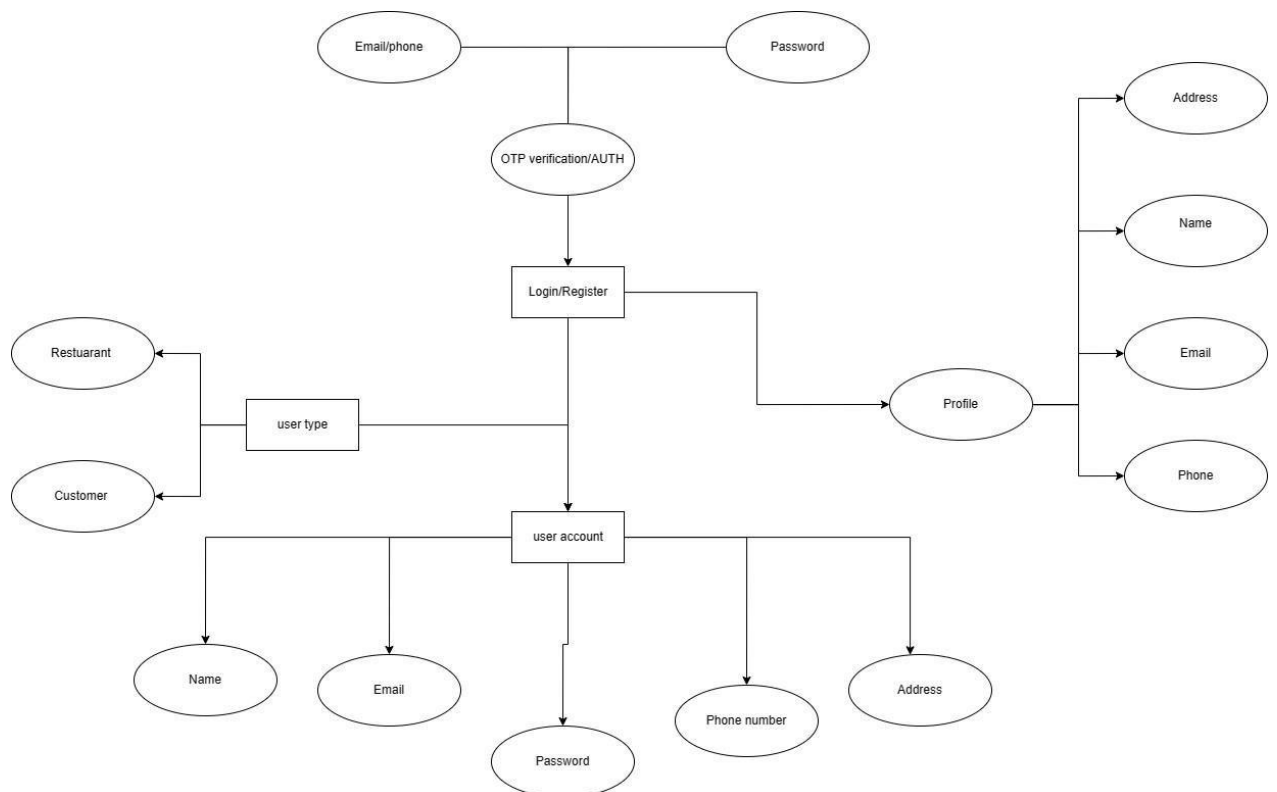


Figure 5 ERD UM





### 5.3. Listings Management System (LM)

#### SRS

| Req Code. | Req. Desc                                                                                                | Testing ID |
|-----------|----------------------------------------------------------------------------------------------------------|------------|
| LM-F-1    | The system should allow the Restaurants to add the foods to the menu list.                               | TC-09      |
| LM-NF-1.1 | When adding item s to the list, they should be able to add photos and update information about the item. | TC-10      |
| LM-F-2    | Restaurants should be able to delete their items.                                                        | TC-11      |
| UM-F-3    | Customers should be able to view all the items in the menu.                                              | TC-12      |

|          |                                                                                |       |
|----------|--------------------------------------------------------------------------------|-------|
| UM-F-3.1 | The customer should be able to click on the listings and view the information. | TC-13 |
| UM-F-4   | The admin should be able to delete items.                                      | TC-14 |

## Activity Diagram

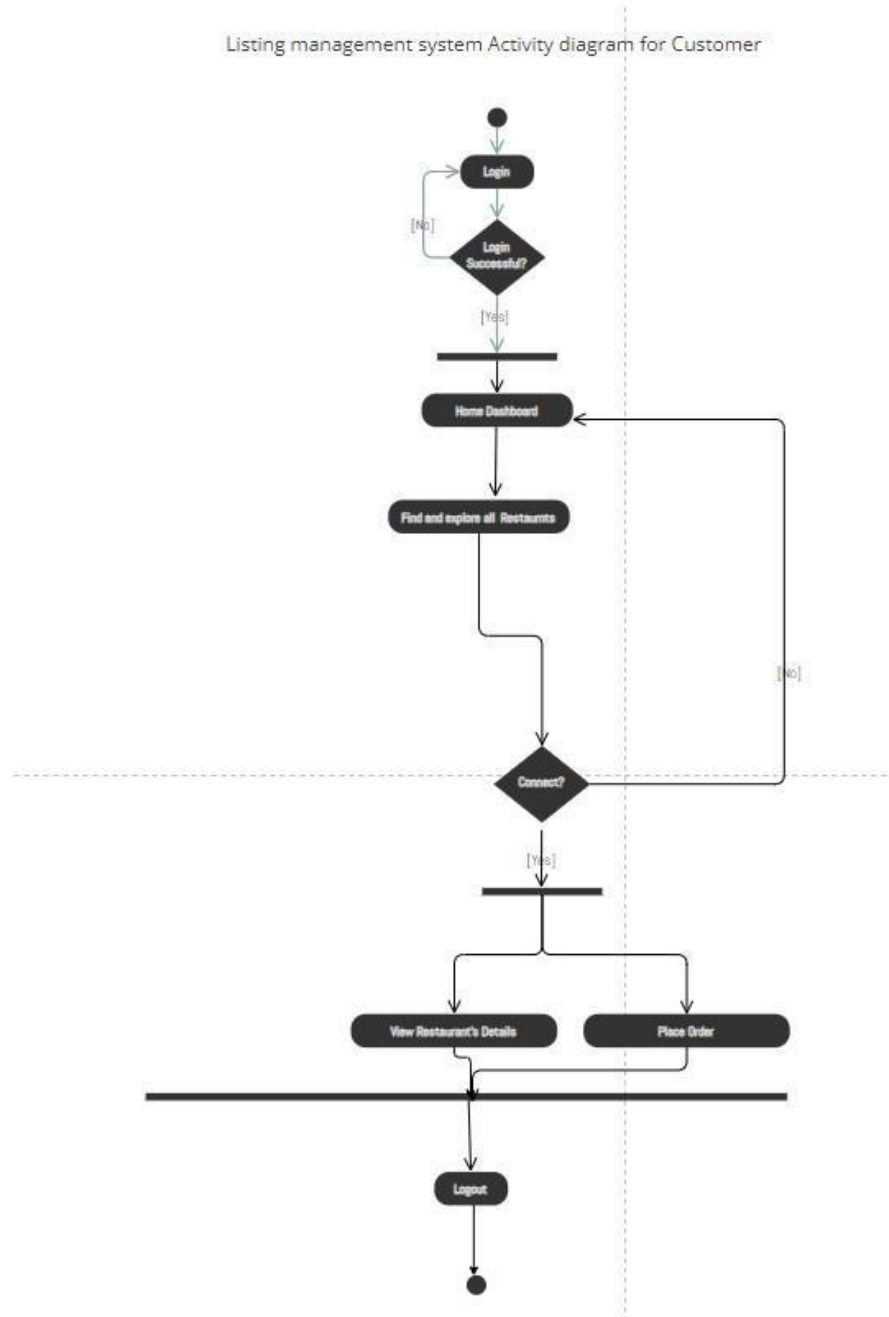


Figure 7 Activity Diagram for Customer LM

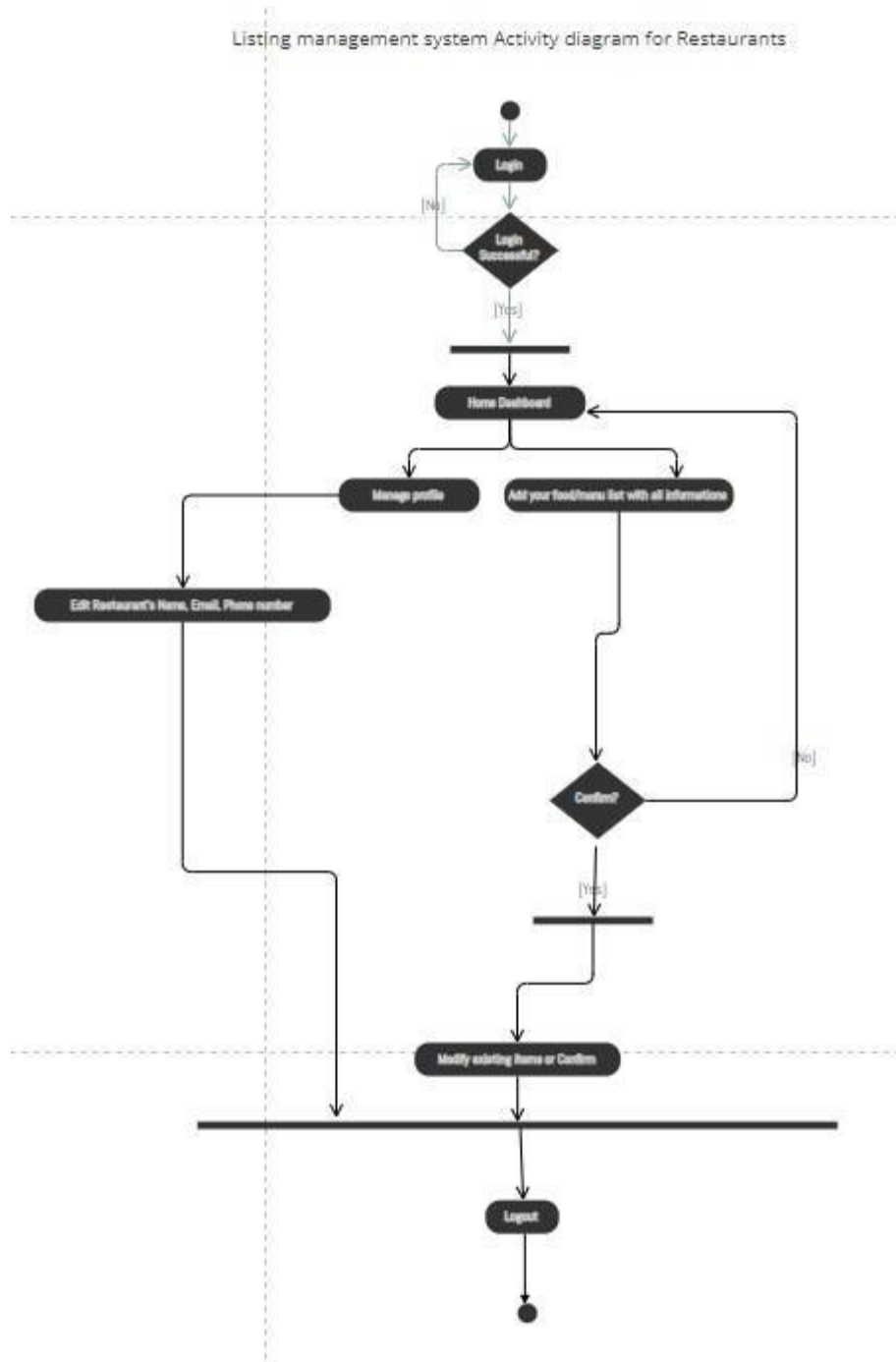


Figure 8 Activity Diagram for Restaurants LM

## Use Case Diagram

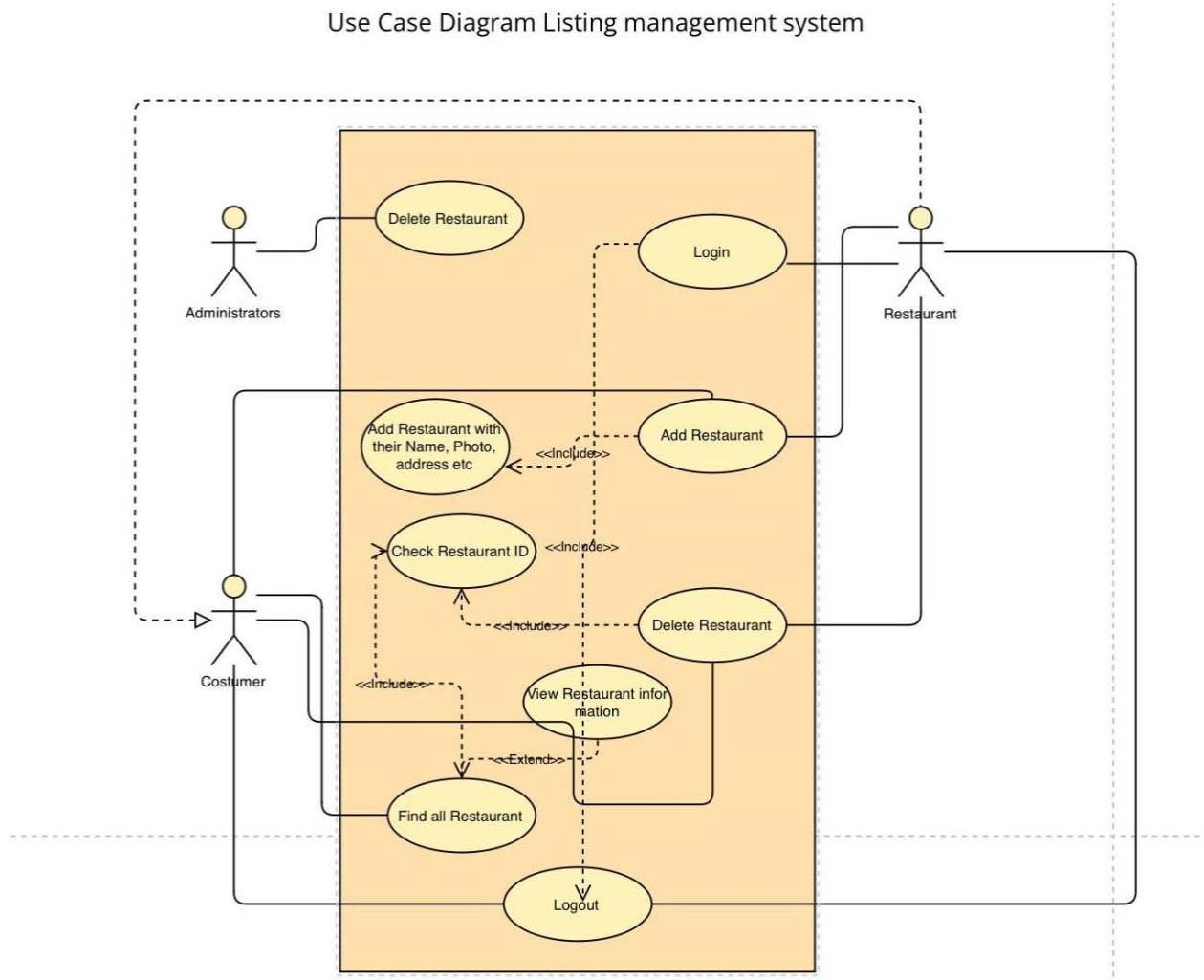


Figure 9 Use Case LM

## Entity Relation Diagram (ERD)

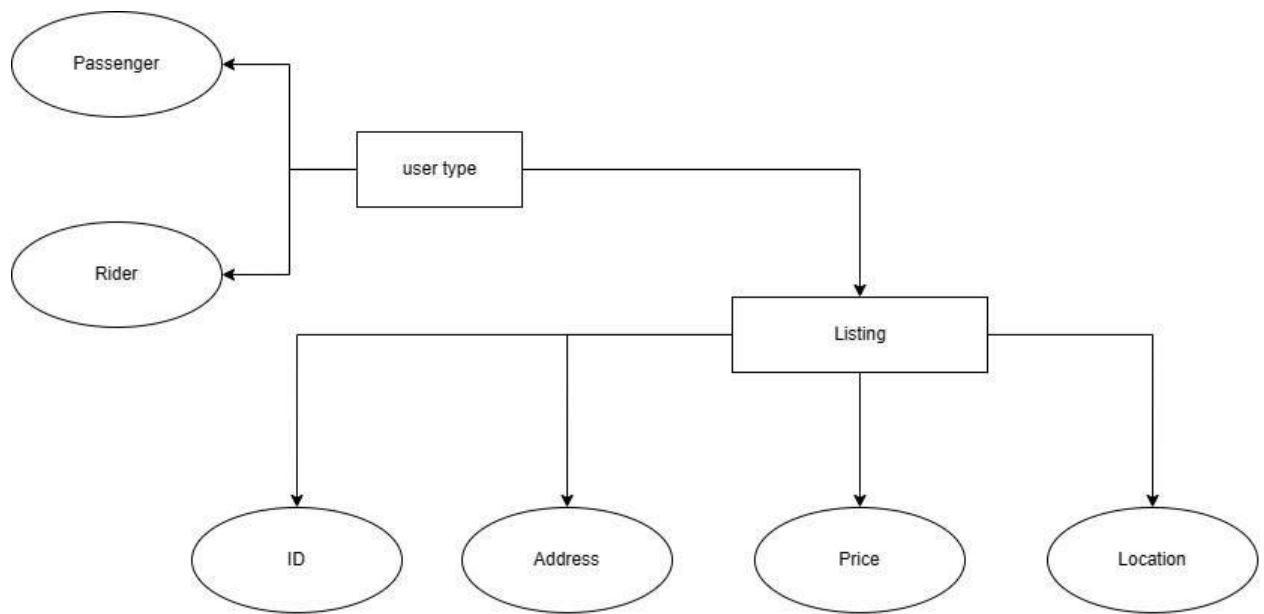


Figure 10 ERD LM

## Sequence Diagram

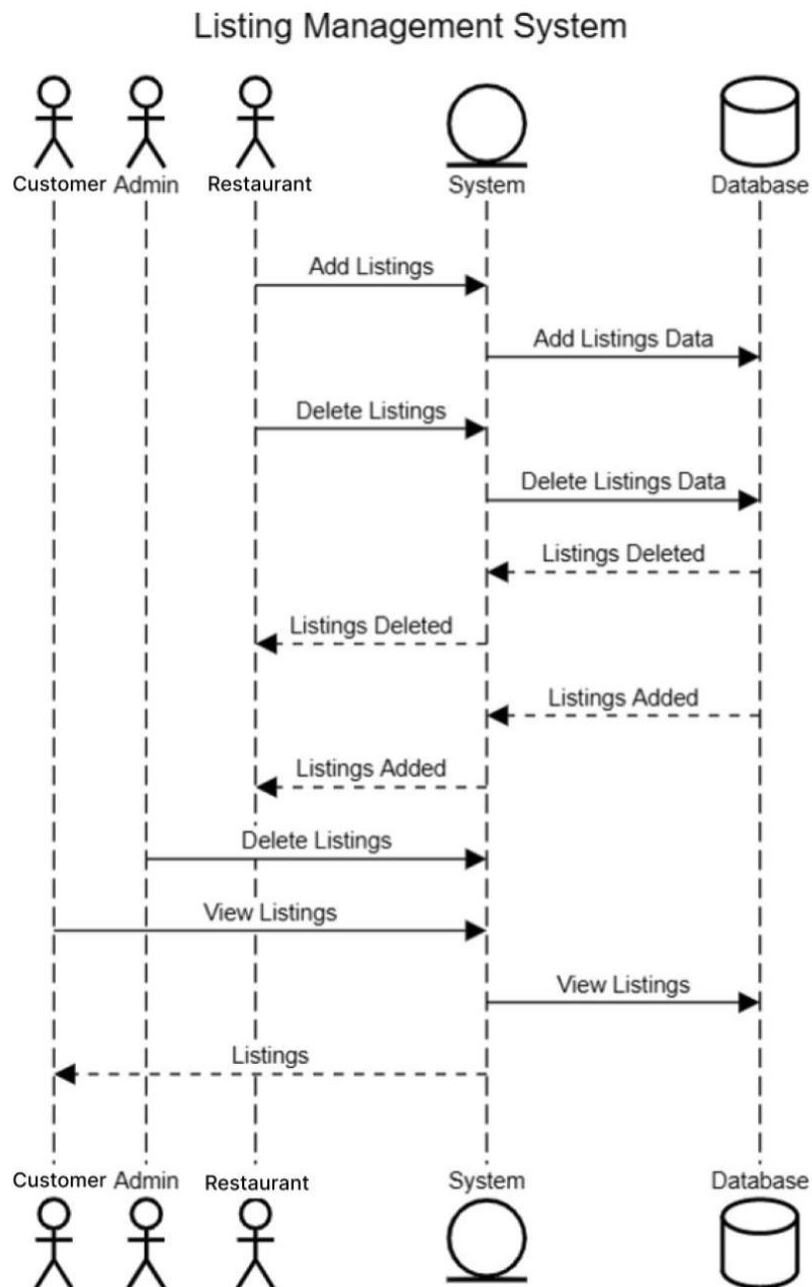


Figure 11 Sequence Diagram LM



## 5.4. Order Management System (OM)

### SRS

| Req Code. | Req. Desc                                                                               | Testing ID |
|-----------|-----------------------------------------------------------------------------------------|------------|
| OM-F-1    | Customers should be able to order the foods listed in the menu.                         | TC-15      |
| OM-F-2    | Restaurants should be able to accept or decline the orders.                             | TC-16      |
| OM-F-3    | Customers can leave feedback for the service of the restaurants or the food experience. | TC-17      |
| OM-F-4    | Restaurants must accept the orders in order to deliver foods.                           | TC-18      |

## Activity Diagram

Order management system Activity diagram for Customers

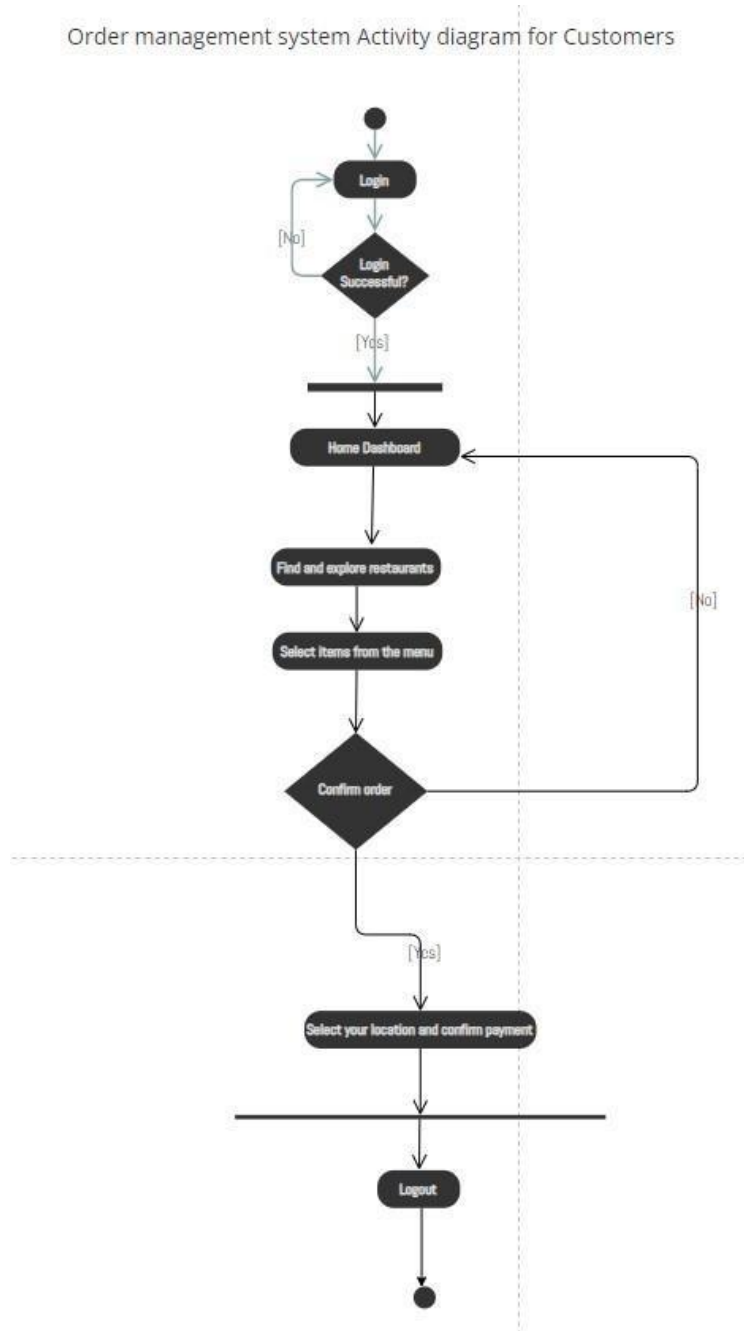


Figure 12 Activity Diagram customer OM

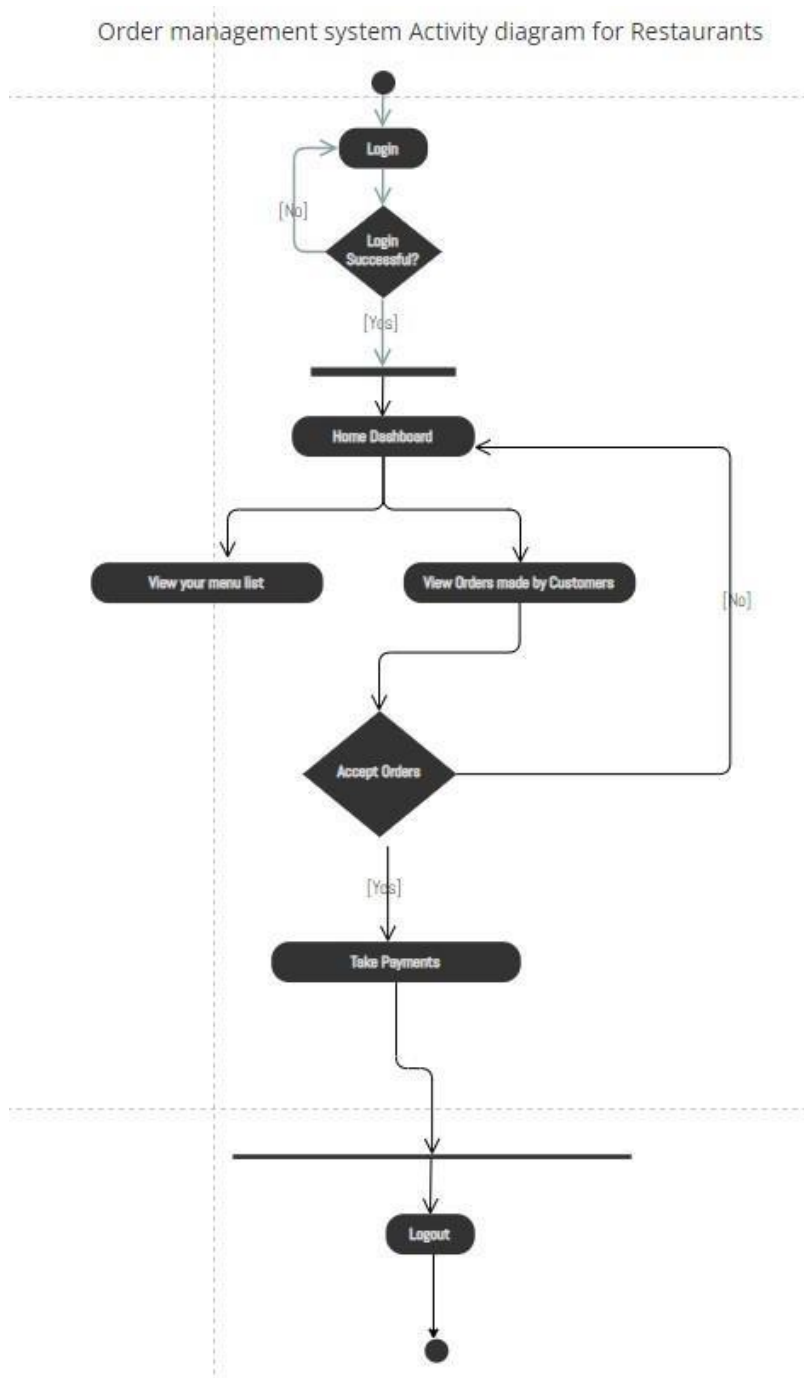
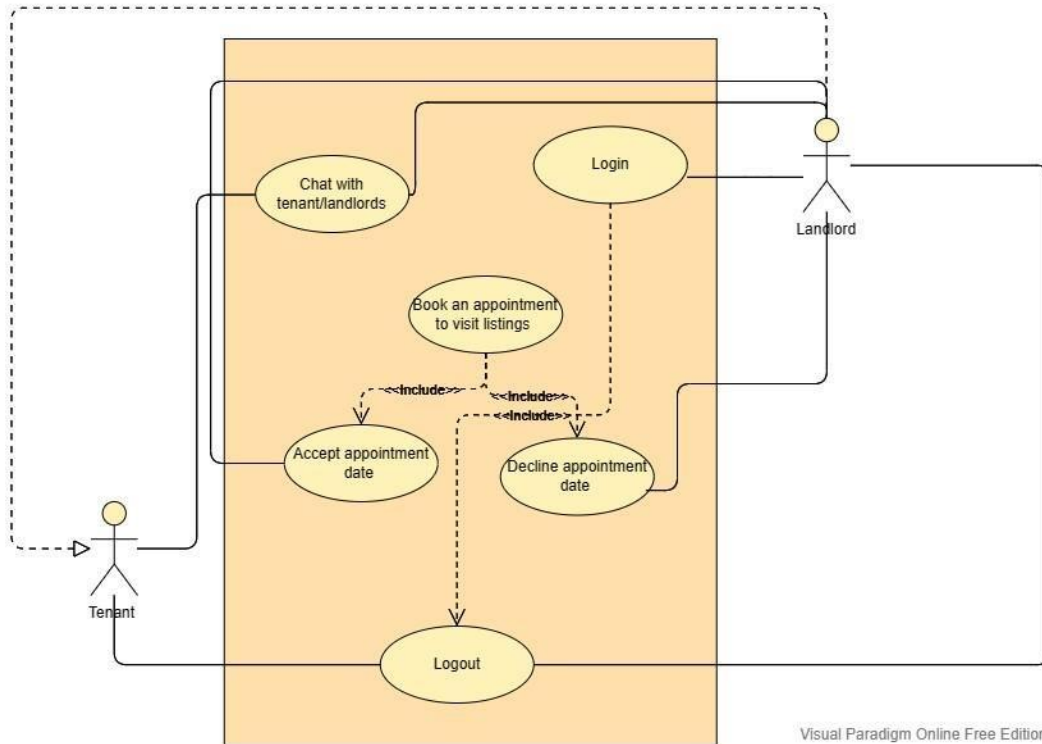


Figure 13 Activity Diagram restaurants OM

## Use Case Diagram

Visual Paradigm Online Free Edition

### Use Case Diagram Booking management system



Visual Paradigm Online Free Edition

Figure 14 Use Case Diagram OM

## Entity Relation Diagram (ERD)

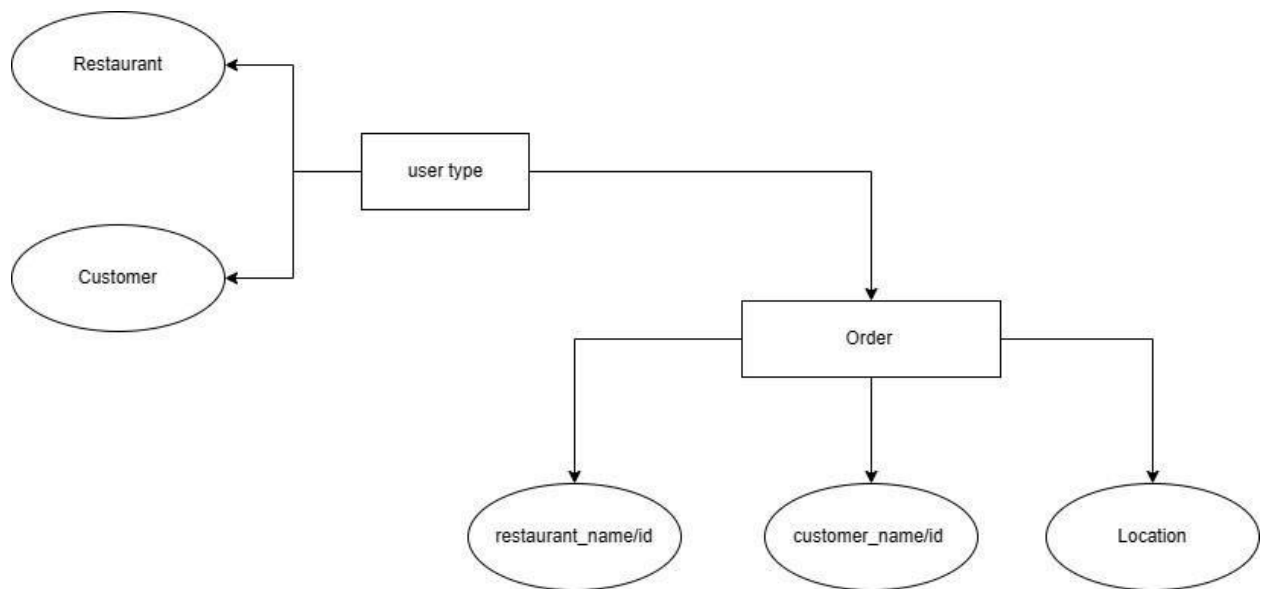


Figure 15 ERD OM

## 5.5. Restaurant Management System (RM)

### SRS

| Req Code. | Req. Desc                                                                          | Testing ID |
|-----------|------------------------------------------------------------------------------------|------------|
| RM-F-1    | The app should allow customers to order foods                                      | TC-19      |
| RM-F-2    | The app should allow restaurants to accept orders from customers easily.           | TC-20      |
| RM-F-3    | After the order is completed and accepted, the customer should get a notification. | TC-21      |
| RM-F-4    | The app should generate the bills for the order of customers.                      | TC-22      |

## Activity Diagram

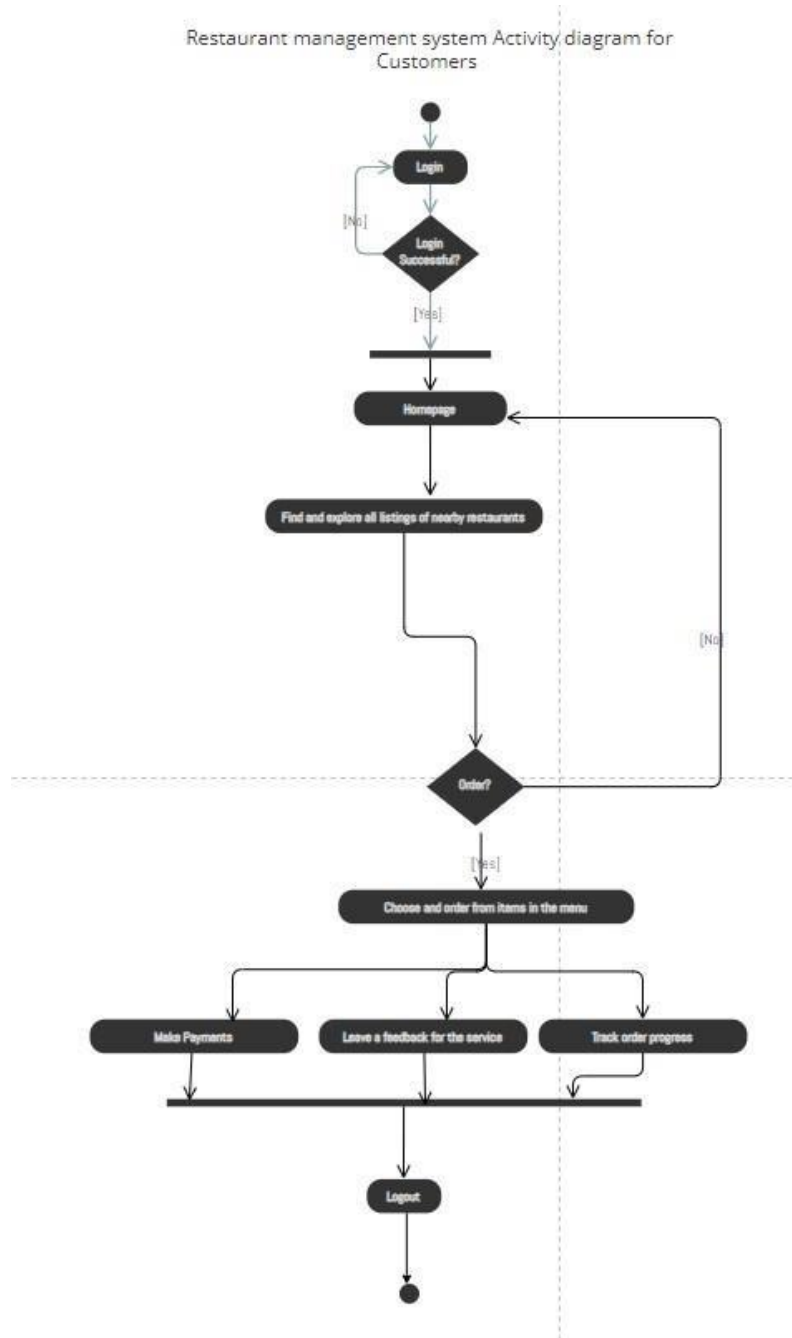


Figure 16 Activity Diagram Customer RM

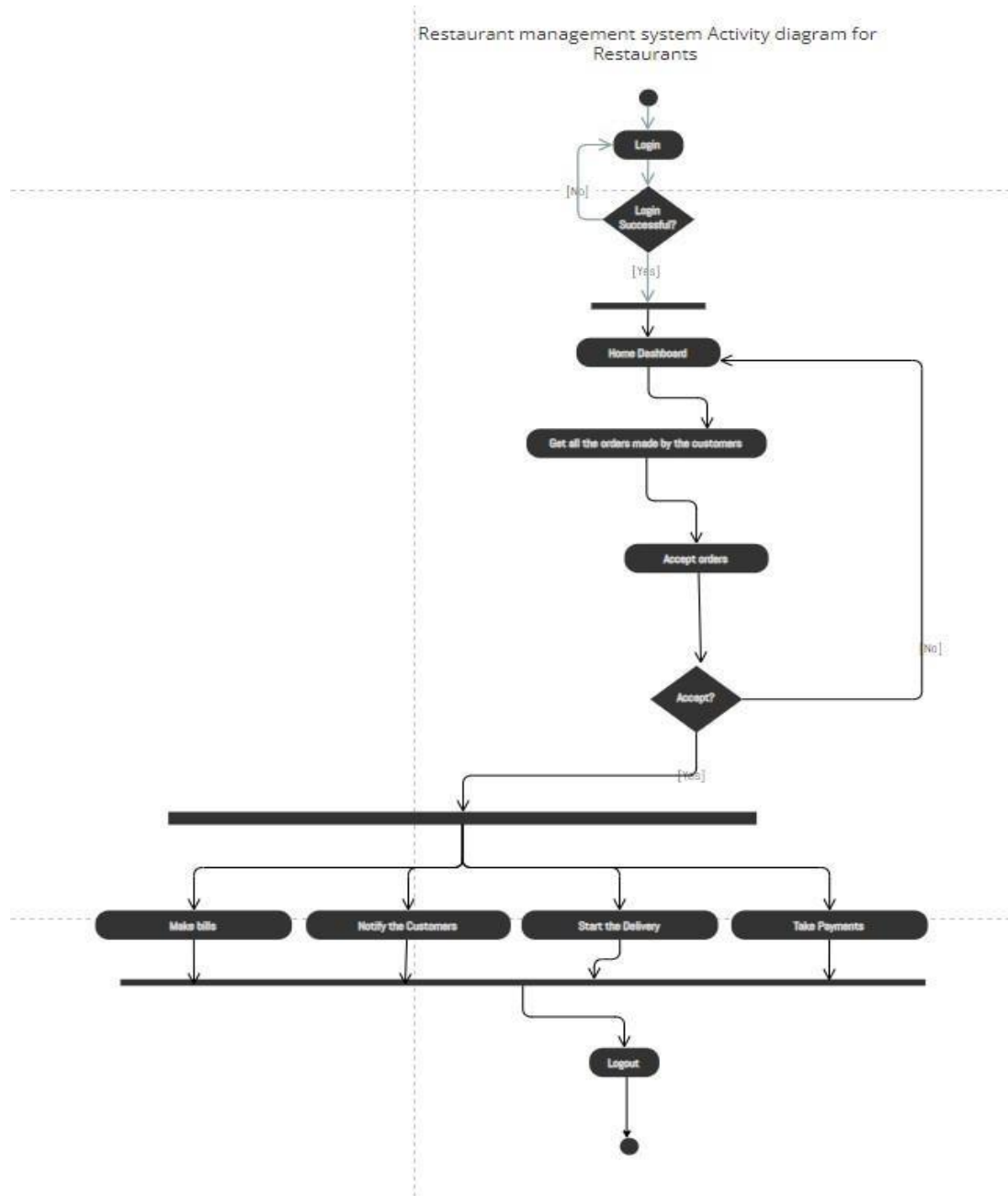




Figure 17 Activity Diagram Restaurants RM

## Use Case Diagram

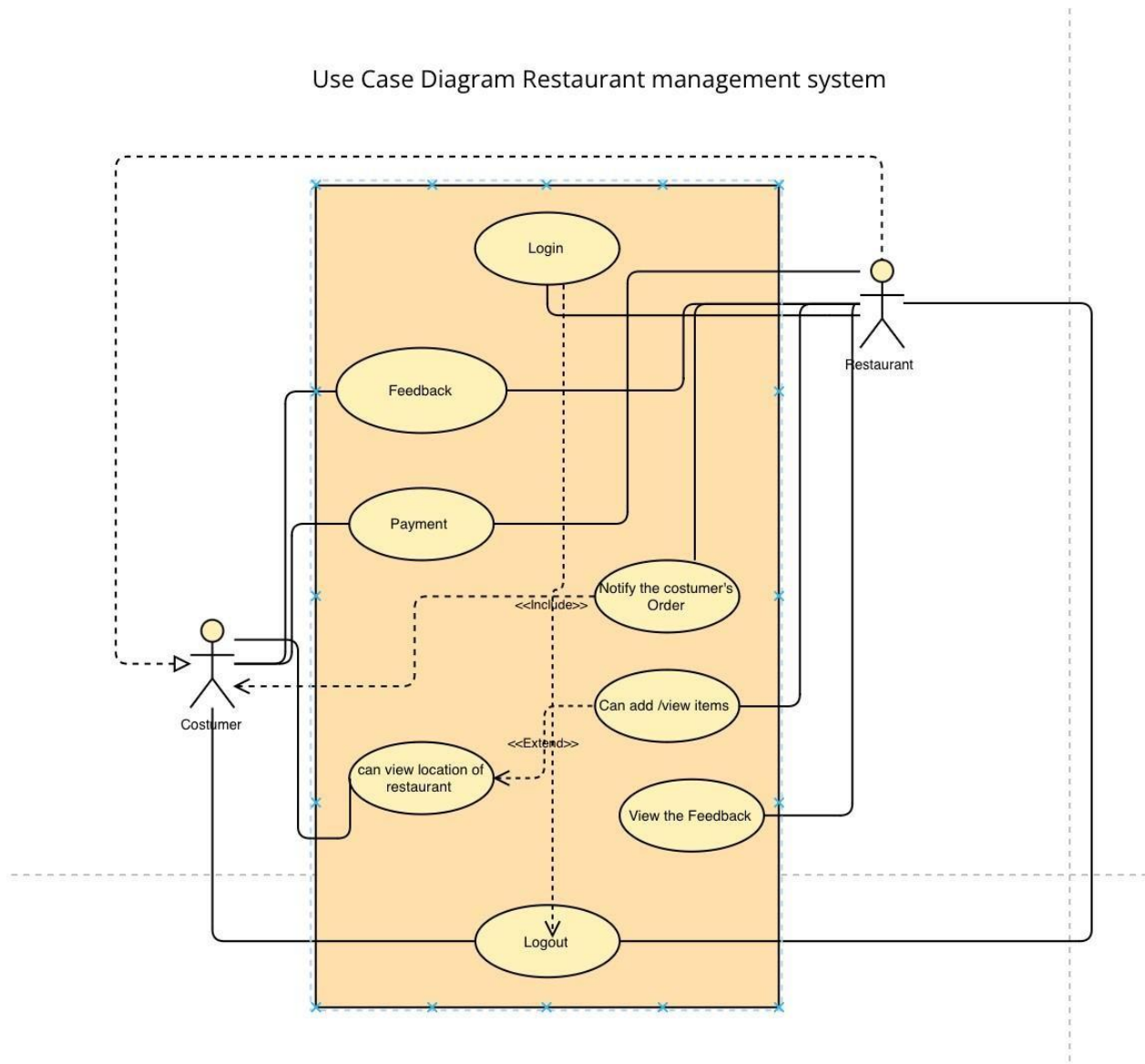


Figure 18 Use Case Diagram Restaurant RM.

## Entity Relation Diagram (ERD)

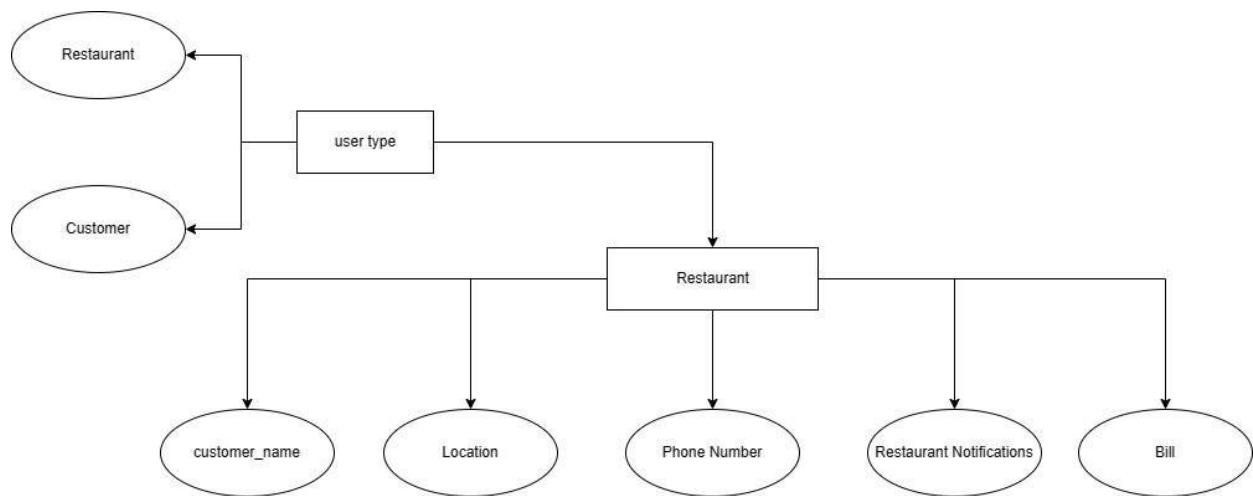


Figure 19 ERD RM

## Testing Table

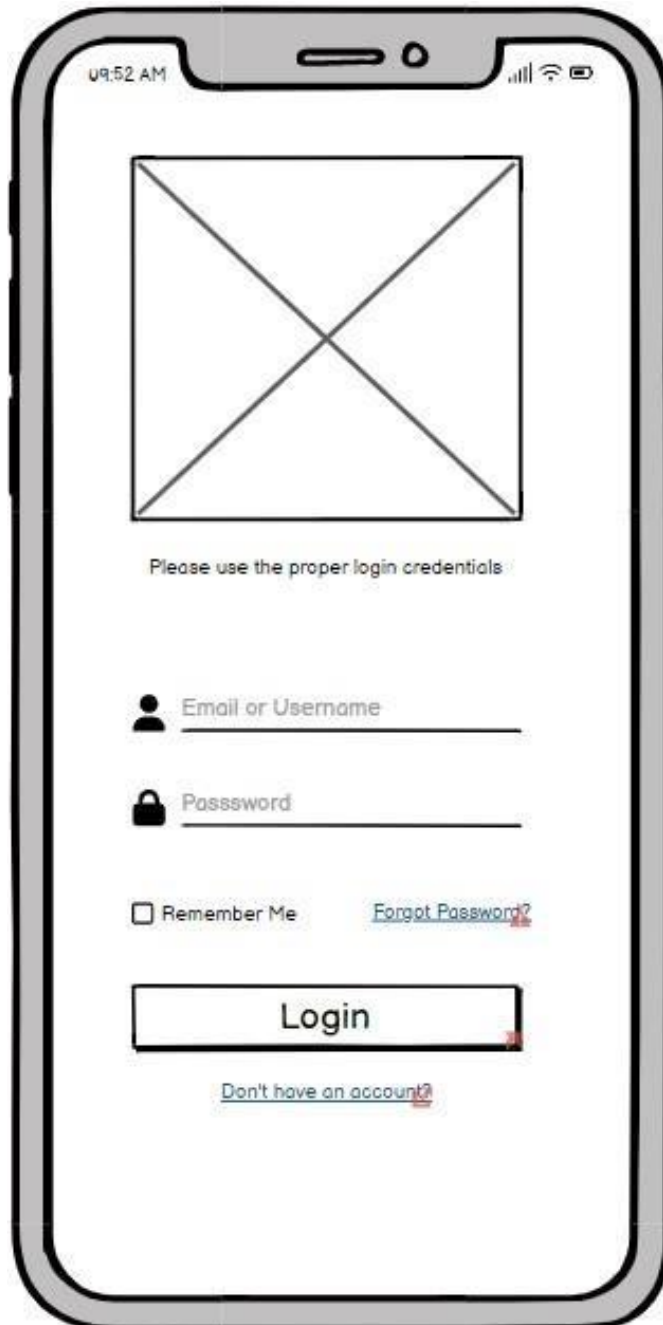
| TC-ID | Testing Name          | SRS-ID        | Expected                                                                                    | Actual                                         | Result<br>Pass/Fail |
|-------|-----------------------|---------------|---------------------------------------------------------------------------------------------|------------------------------------------------|---------------------|
| TC-01 | Register              | UM-F-1        | Successful registration<br>With Email/phone no<br>Also, Google/Phone<br>number registration | Login/Registration<br>n Successfully           | Pass                |
| TC-02 | Password              | UM-<br>NF-1.1 | The password must be<br>combination of<br>alphabets/numbers and<br>special characters       | System allowed<br>password of<br>numbers only. | Pass                |
| TC-03 | Admin delete<br>users | UM-F-2        | The admin must be<br>able to delete users.                                                  | User deleted<br>successfully.                  | Pass                |
| TC-04 | User profile          | UM-F-3        | The users should have<br>their profile with their<br>information.                           | System<br>displayed profile<br>successfully.   | Pass                |

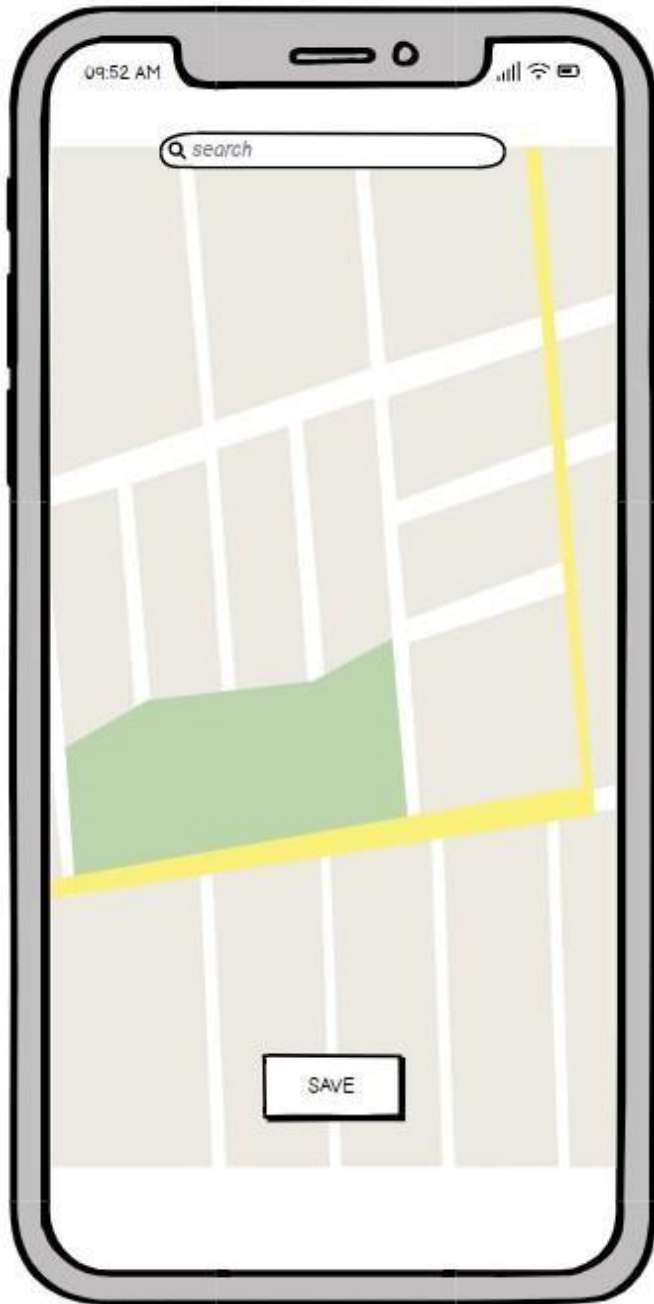
|       |                          |        |                                                             |                                        |      |
|-------|--------------------------|--------|-------------------------------------------------------------|----------------------------------------|------|
| TC-06 | Restaurants remove items | UM-F-5 | The restaurants must be able to delete items from the menu. | Restaurant removed items successfully. | Pass |
|-------|--------------------------|--------|-------------------------------------------------------------|----------------------------------------|------|

|       |                           |        |                                                                            |                               |      |
|-------|---------------------------|--------|----------------------------------------------------------------------------|-------------------------------|------|
| TC-08 | Leave Feedback            | UM-F-7 | The app should allow customers to leave feedback.                          | Comments worked successfully. | Pass |
| TC-09 | Add listings              | LM-F-1 | Restaurants should be able to add items to the menu.                       | Listing added.                | Pass |
| TC-12 | View listings             | LM-F-3 | App should allow customers to view all the items in the restaurant's menu. | Listing view.                 | Pass |
| TC-14 | Delete listings via admin | LM-F-4 | The app should allow the admin to delete and view the items.               | Items deleted                 | Pass |

|       |                   |        |                                                                                                   |                        |      |
|-------|-------------------|--------|---------------------------------------------------------------------------------------------------|------------------------|------|
| TC-15 | Order Foods.      | OM-F-1 | Customers should be able to order foods from the menu.                                            | Ordered successfully.  | Pass |
| TC-09 | Reject Orders.    | OM-F-2 | Restaurant should be able to decline the orders.                                                  | Successfully Rejected. | Pass |
| TC-21 | Notify Customers. | RM-F-3 | After the order is successfully completed, Customers should be notified about the order progress. | Customers Notified.    | Pass |
| TC-22 | Generate Bills.   | RM-F-4 | The app should generate bills based on the customer's order.                                      | Bills generated.       | Pass |

## 5.6. Wireframe





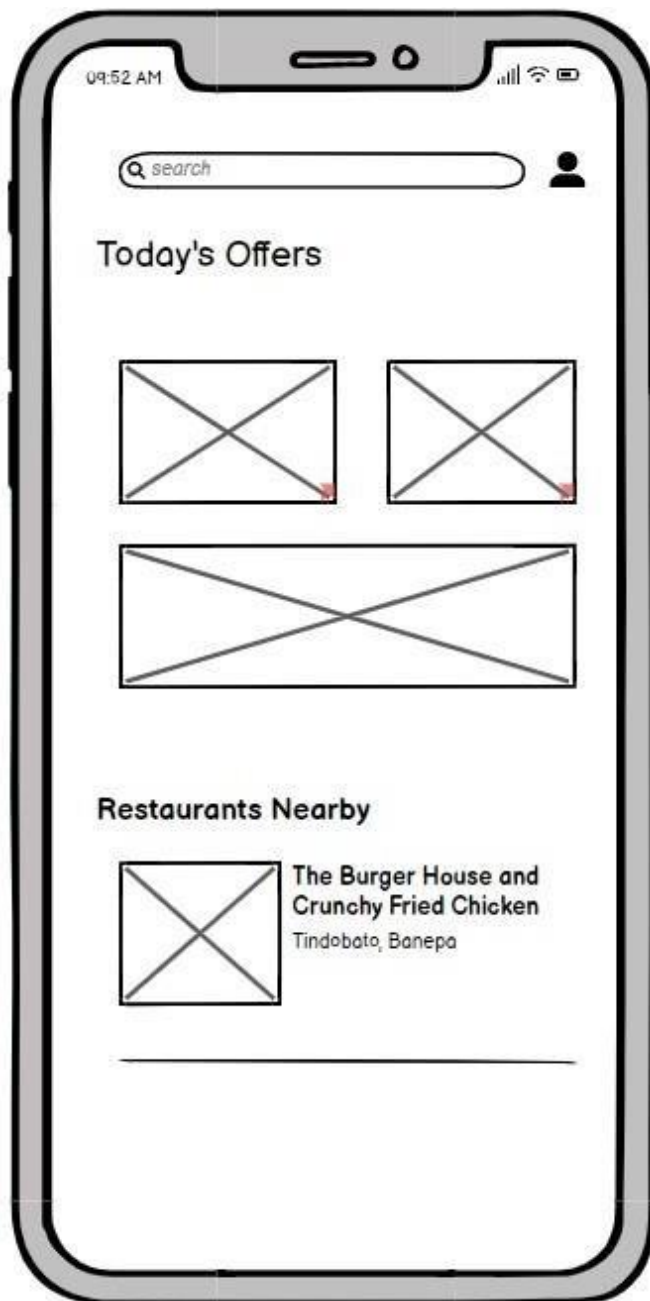








Figure 23 Figma 1 (Loading Screen)

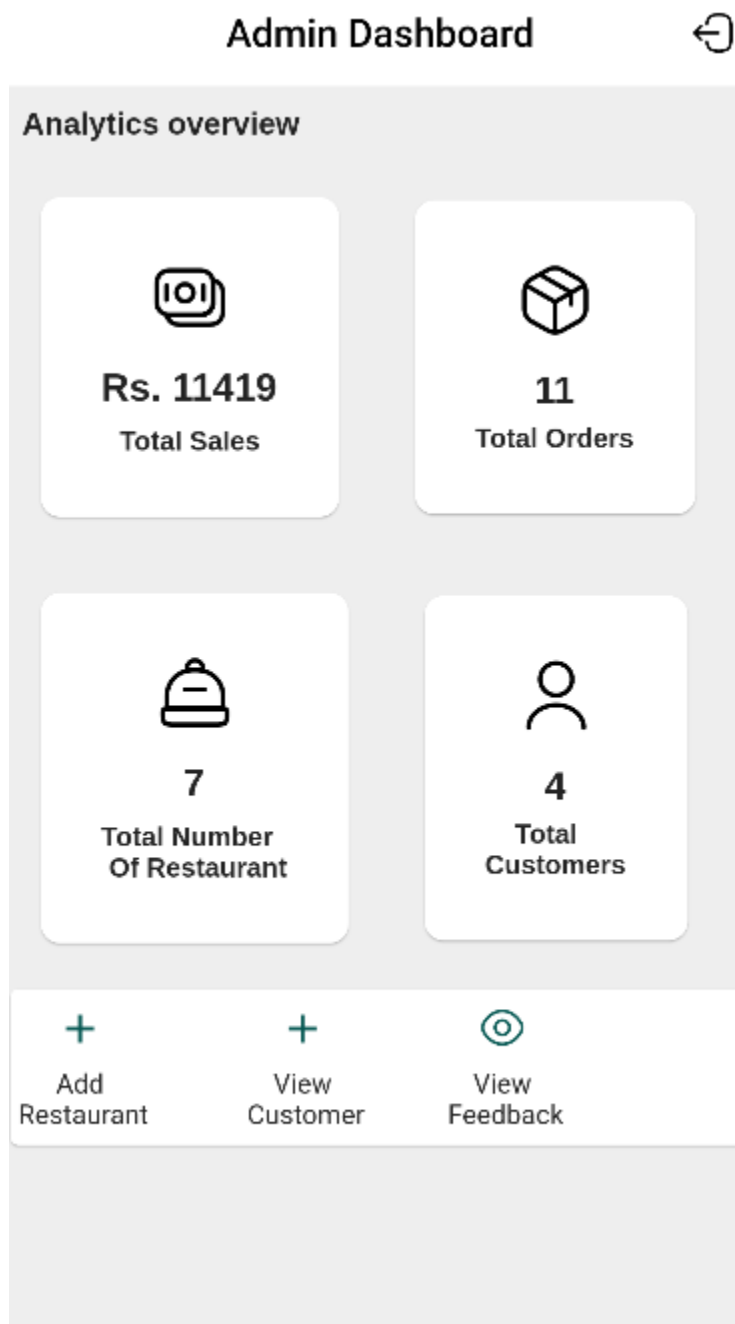
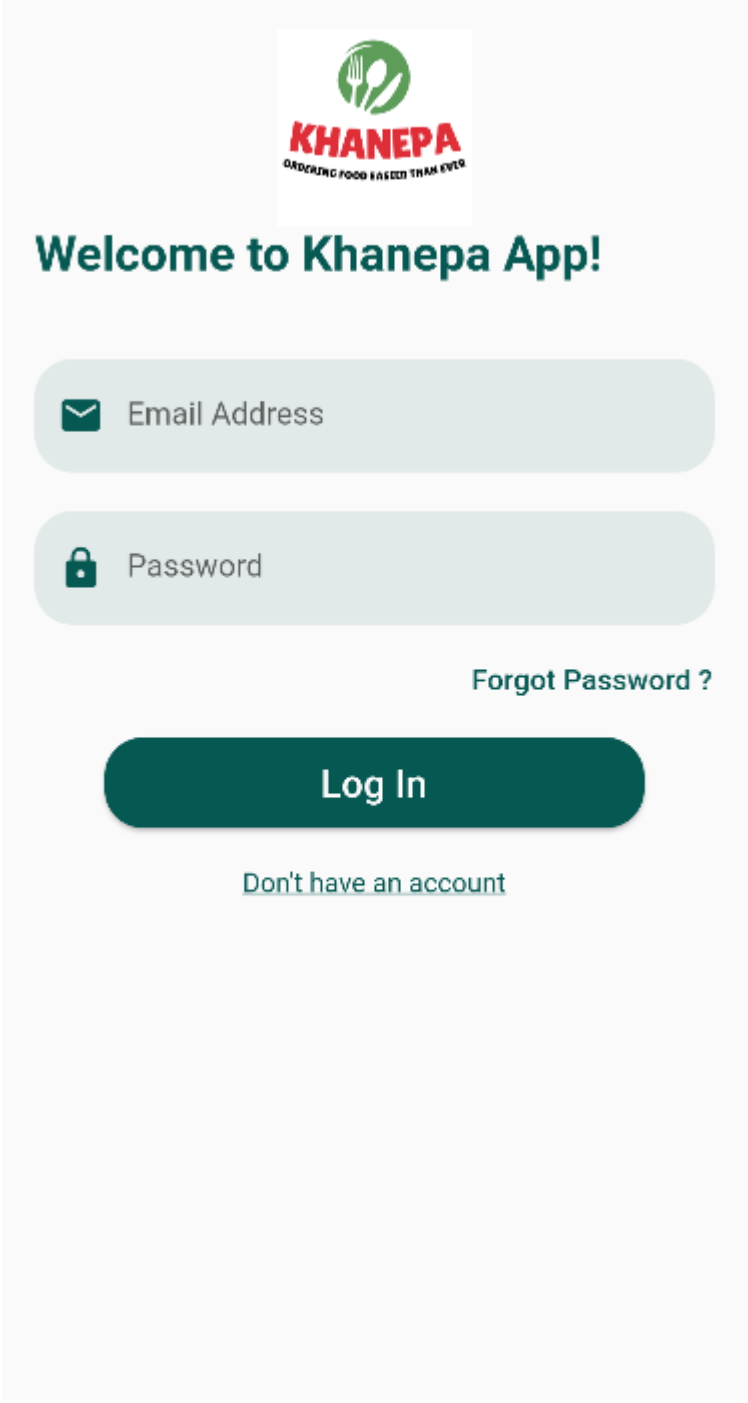


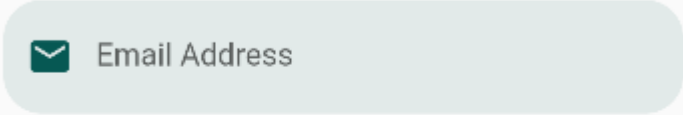
Figure 24 Figma 2

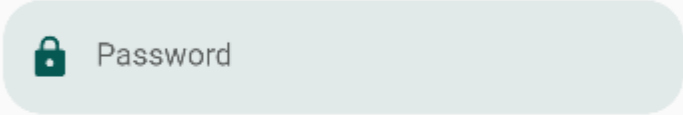


The image shows a mobile app login screen for 'Khanepa'. At the top center is the Khanepa logo, which consists of a green circle containing a white fork and spoon icon, with the word 'KHANEPA' in bold red letters below it, and the tagline 'ORDERING FOOD EASIER THAN EVER' in smaller black letters. Below the logo, the text 'Welcome to Khanepa App!' is displayed in a bold, dark green font. There are two input fields: the first is labeled 'Email Address' with a green envelope icon to its left, and the second is labeled 'Password' with a green padlock icon to its left. Both fields have a light gray background and rounded corners. Below the password field, there is a link that says 'Forgot Password ?' in a dark green font. A large, dark green button with rounded corners and the text 'Log In' in white is positioned below the link. At the bottom, there is a link that says 'Don't have an account' in a dark green font.



## Welcome to Khanepa App!





[Forgot Password ?](#)

[Log In](#)

[Don't have an account](#)

Figure 25 Figma 3

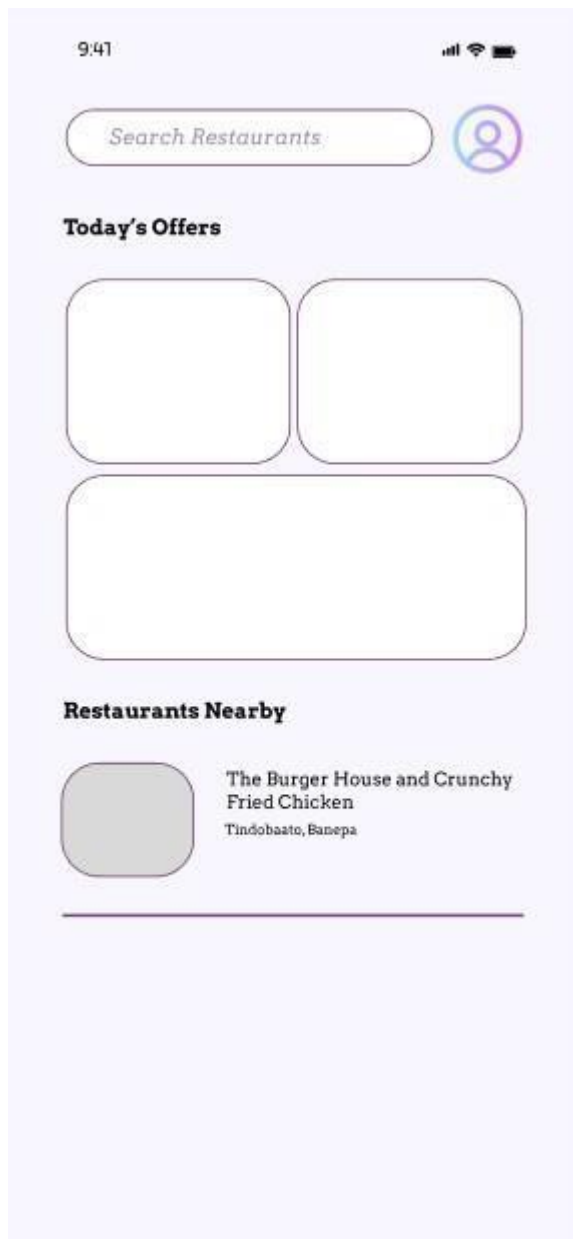


Figure 26 Figma 4

### 5.6.1. WBS

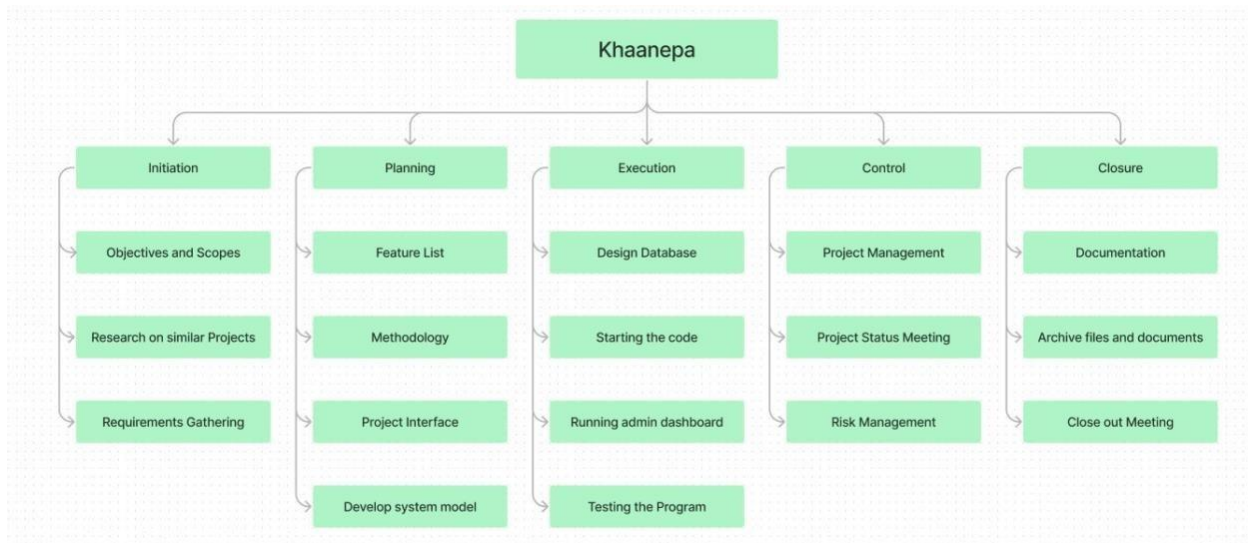


Figure 1 WBS

### 5.6.2. Initial Gantt Chart



Figure 2 Initial Gantt Chart

## 5. Conclusion

The main goal of this project was to create an effective and user-friendly mobile application that makes ordering food and finding restaurants easier. In order to ensure the best outcome, in-depth investigation and analysis were carried out at every stage of the development process. The app has a clean, user-friendly interface that prioritizes simplicity to improve user comprehension and usability. The thorough report contains meticulously documented summaries of all the research discoveries and realizations made throughout the development process. The actual application has undergone extensive development with a focus on building a platform that is not only simple to use but also completes tasks quickly and effectively. User needs and preferences have received special consideration, ensuring that each user has a simple and engaging experience.

I am confident that this application will provide significant value to every one of its users, especially in light of the efforts put into the project and how well it complies with user requirements. The app aims to improve the overall dining experience and meet the various needs of its users by streamlining the food ordering process and offering a handy restaurant discovery feature.



## 6. Critical Evaluation

### 6.1 Final Report

The project overall went very smooth. Although I had a lot of difficulties, I managed to build a simple functioning app. I failed 3 times and made 3 different apps before finally making this final functioning app. It was always some small issues that made the project difficult. I managed to get the listed features implemented in the app. Some of the features that I tried adding but could not add were implemented in an alternative manner. Those features include Google Maps API and Khalti API. As an alternative to Google Maps API, I have made it possible for admin to add the restaurant's location in URL. Then, if the customers want to view the location of the restaurant, there is a button dedicated to Google Map which once clicked, will open Google Maps App itself and view the restaurant's Location. For the payment method, I removed the online services as I encountered errors with integrating APIs on the final stages of the project and was afraid that the error would affect the project as a whole.

Although the project didn't quite go according to my initial plan, I was able to make the best of what I could. I found myself lacking time management skills in the initial phases of the project. But over time, I managed to develop my time management skills too. During the development of the project, I was able to get my hands on ample number of resources on the internet, most of it being from the community discussions, YouTube, and many websites that help in solving the issues related to coding. All the sources I used turned out to be very helpful and were used in most cases.

#### Logo Used for this App:



Figure 1: App Logo

## **6.2 Findings**

While I researched the project and other similar systems, I found out that a lot of Food Ordering Apps were not very friendly for all the types/ age group of people. To tackle those kinds of problems, I made the app very simple and easy to use for all age groups with very straight forward UI and features. My App contains very simple but effective features and does not go through other advanced features that might create confusion among users. I used a lot of references from other food ordering apps, not the design and working mechanisms, but the key features. As I was developing a food delivery app, I also did a little research on current food ordering/ delivery apps in my area. I came to know that currently, there are no such apps providing services in my area from which I was even more inspired to work on this project. Inspired by my city name Banepa, I came up with the name of the app 'Khanepa'. I also did a little research on how often people tend to order food online from a restaurant in my area. I found out that people in my area order a lot of food from restaurants on a regular basis via a primary and only option, which is calling the restaurant directly.

## **6.3 Planning**

Talking about what went according to the plan and what did not, almost every feature was implemented in the app, however, I had problems with integrating APIs and was forced to use other alternatives. Alternatives suggested by the supervisor were implemented in the system. The planning process was interrupted by many problems as I had to develop multiple apps before the final app. I kept encountering issues with the integration of APIs which delayed my work progress and even starting the project all over again.

## **6.4 Research and Process**

As said earlier, I used a lot of references and did a lot of research before and during the

development of this app as my knowledge to developing an app, and especially back-end development, was very little. The app is completely made from flutter and firebase. Learning the features of flutter and firebase was much easier after I completed a few courses from YouTube and Coursera. Not to forget, a lot of my queries were also solved with the help of various community discussion platforms like stack overflow. For some of my major issues, our supervisor also helped me to go through other possible alternatives.

## **6.5 Self-Reflection**

As said earlier, the number one thing I found within myself is time management skills. During the initial stage of the project, I often found myself not completing the goals I would have set for the given time. Upon completion of the project, I developed and improved my skills in flutter and firebase. As I was just a beginner in Flutter, I had a lot of difficulties learning but overtime, I did learn a lot about it and now, I find myself comfortable with most of the features of flutter. About Firebase, it was much more different than I thought it would be. It was more challenging than I had imagined. Talking about the most difficult part, I would say the back end was very challenging as I had never worked in back-end designing of any project before. Not only in the development of the project itself, but this project has also helped me in developing myself as a person too. I developed my research skills, understanding of the researched data, how to know if the data are relevant, and how to implement and use the available resources to the full potential.

## 7. Evidence of Project Management

### 7.1 Logsheet

Faculty of Science and Engineering  
School of Mathematics and Computer Science



UNIVERSITY OF  
WOLVERHAMPTON



| PROJECT MANAGEMENT LOG                                                                                                                                                                                                                                                                                              |                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| First Name: <u>Ashim</u>                                                                                                                                                                                                                                                                                            | Surname: <u>Bhochhibhoya</u>    |
| Student Number: <u>2059144</u>                                                                                                                                                                                                                                                                                      | Supervisor: <u>Anil Poudyal</u> |
| Project Title: <u>Khanepa</u>                                                                                                                                                                                                                                                                                       | Month: <u>November</u>          |
| What have you done since the last meeting                                                                                                                                                                                                                                                                           |                                 |
| <ol style="list-style-type: none"> <li>1. Collected requirements for the project.</li> <li>2. <u>Researched for</u> the relevant tools and technologies for the development of the project.</li> <li>3. Made Gannt Chart for the project.</li> <li>4. <u>Researched on</u> similar applications.</li> </ol>         |                                 |
| What do you aim to complete before the next meeting                                                                                                                                                                                                                                                                 |                                 |
| <ol style="list-style-type: none"> <li>1. Complete Literature review draft</li> <li>2. Improve the proposal draft for the <u>project</u></li> </ol>                                                                                                                                                                 |                                 |
| Supervisor comments                                                                                                                                                                                                                                                                                                 |                                 |
| <ol style="list-style-type: none"> <li>1. <u>Introduction</u> is relevant and well explained.</li> <li>2. Research on similar <u>application</u> is relevant. Research on algorithm and analysis has been mentioned.</li> <li>3. Report structure and format not followed according to the instructions.</li> </ol> |                                 |

☐

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Supervisor Signature: 

Date: 13<sup>th</sup> November, 2022

Date: 13<sup>th</sup> November, 2022

Figure 2: Logsheet 1

Faculty of Science and Engineering  
School of Mathematics and Computer Science



| PROJECT MANAGEMENT LOG                                                                                                                                      |                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| First Name: Ashim                                                                                                                                           | Surname: Bhochhibhoya    |
| Student Number: 2059144                                                                                                                                     | Supervisor: Anil Poudyal |
| Project Title: <u>Khanepa</u>                                                                                                                               | Month: December          |
| What have you done since the last meeting                                                                                                                   |                          |
| 1. Literature Review<br>2. proposal draft for the project                                                                                                   |                          |
| What do you aim to complete before the next meeting                                                                                                         |                          |
| 1. Work on Artefact Design<br>2. Create login page<br>3. Research on Firebase and Flutter                                                                   |                          |
| Supervisor comments                                                                                                                                         |                          |
| 1. The selected methodology should inform the development of the Gantt Chart. There should be more study on Scrum.<br><br>2. Feedback of literature review. |                          |

We confirm that the information given in this form is true, complete, and accurate.

Student Signature: 

Date: 2022/12/7

Supervisor Signature: 

Date: 2022/12/7

Figure 3: Logsheets 2

Faculty of Science and Engineering  
School of Mathematics and Computer Science



| PROJECT MANAGEMENT LOG                                                                                                                                           |                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| First Name: Ashim                                                                                                                                                | Surname: Bhochhibhoya    |
| Student Number: 2059144                                                                                                                                          | Supervisor: Anil goudyal |
| Project Title: <del>Khanna</del>                                                                                                                                 | Month: January           |
| What have you done since the last meeting                                                                                                                        |                          |
| <ol style="list-style-type: none"> <li>1. Worked on Login Page.</li> <li>2. Worked on artefact design.</li> <li>3. Researched on Firebase and Flutter</li> </ol> |                          |
| What do you aim to complete before the next meeting                                                                                                              |                          |
| <ol style="list-style-type: none"> <li>1. Work on Home Page.</li> <li>2. Make Activity Diagram</li> </ol>                                                        |                          |
| Supervisor comments                                                                                                                                              |                          |
| <ol style="list-style-type: none"> <li>1. Make a draft for the artifact design.</li> </ol>                                                                       |                          |

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 2023/1/4

Supervisor Signature: 

Date: 2023/1/4

Figure 4: Logsheets 3

| PROJECT MANAGEMENT LOG                                                                                                                                                      |                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| First Name: Ashim                                                                                                                                                           | Surname: Bhochhibhoya    |
| Student Number: 2059144                                                                                                                                                     | Supervisor: Anil Poudyal |
| Project Title: <del>Khanepa</del>                                                                                                                                           | Month: January           |
| What have you done since the last meeting                                                                                                                                   |                          |
| <ol style="list-style-type: none"> <li>1. Worked on Home page design.</li> <li>2. Worked on Use case diagram of the application.</li> </ol>                                 |                          |
| What do you aim to complete before the next meeting                                                                                                                         |                          |
| <ol style="list-style-type: none"> <li>1. Prepare product backlog</li> <li>2. Make class diagram</li> <li>3. Make a Dashboard which includes profile, list, etc.</li> </ol> |                          |
| Supervisor comments                                                                                                                                                         |                          |
| <ol style="list-style-type: none"> <li>1. Complete your activity diagram</li> <li>2. Create User app UI.</li> </ol>                                                         |                          |

We confirm that the information given in this form is true, complete and accurate.

Student Signature:



Date: 2023/1/11

Supervisor Signature:



Date: 2023/4/9

Figure 5: Logsheets 4

| PROJECT MANAGEMENT LOG                                           |                              |
|------------------------------------------------------------------|------------------------------|
| First Name: <u>Ashim</u>                                         | Surname: <u>Bhochhibhoya</u> |
| Student Number: 2059144                                          | Supervisor: Anil Poudyal     |
| Project Title: <u>Khanepa</u>                                    | Month: February              |
| What have you done since the last meeting                        |                              |
| 3. Added Restaurant Dashboard<br>4. Planned to add another page. |                              |
| What do you aim to complete before the next meeting              |                              |
| 3. Add the feedback page.<br>4. Improve dashboard UI.            |                              |
| Supervisor comments                                              |                              |
| Fix the error on the dashboard panel.                            |                              |

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 13<sup>th</sup> February, 2023

Supervisor Signature: 

Date: 13<sup>th</sup> February, 2023

Figure 6: Logsheet 6





| PROJECT MANAGEMENT LOG                                           |                          |
|------------------------------------------------------------------|--------------------------|
| First Name: Ashim                                                | Surname: Bhochhibhoya    |
| Student Number: 2059144                                          | Supervisor: Anil Poudyal |
| Project Title: Khanepa                                           | Month: February          |
| What have you done since the last meeting                        |                          |
| 3. Added Restaurant Dashboard<br>4. Planned to add another page. |                          |
| What do you aim to complete before the next meeting              |                          |
| 3. Add the feedback page.<br>4. Improve dashboard UI.            |                          |
| Supervisor comments                                              |                          |
| Fix the error on the dashboard panel.                            |                          |

We confirm that the information given in this form is true, complete and accurate.

Student Signature:



Date: 13<sup>th</sup> February, 2023

Supervisor Signature:



Date: 13<sup>th</sup> February, 2023

Figure 7: Logsheet 7

| PROJECT MANAGEMENT LOG                                                                         |                          |
|------------------------------------------------------------------------------------------------|--------------------------|
| First Name: Ashim                                                                              | Surname: Bhochhibhoya    |
| Student Number: 2059144                                                                        | Supervisor: Anil Poudyal |
| Project Title: Khanepa                                                                         | Month: March             |
| What have you done since the last meeting                                                      |                          |
| 1. Completed the feedback option.<br>2. Completed adding restaurants via <a href="#">admin</a> |                          |
| What do you aim to complete before the next meeting                                            |                          |
| 1. Complete draft report                                                                       |                          |
| Supervisor comments                                                                            |                          |
| 1. Remove bug from admin page                                                                  |                          |

We confirm that the information given in this form is true, ~~complete~~ and accurate.

Student Signature:



Date: 2<sup>nd</sup> March 2023

Supervisor Signature:



Date: 2<sup>nd</sup> March, 2023

Figure 8: Logsheet 8

Faculty of Science and Engineering  
School of Mathematics and Computer Science



| PROJECT MANAGEMENT LOG                              |                          |
|-----------------------------------------------------|--------------------------|
| First Name: Ashim                                   | Surname: Bhochhibhoya    |
| Student Number: 2059144                             | Supervisor: Anil Poudyal |
| Project Title: Khanepa                              | Month: March             |
| What have you done since the last meeting           |                          |
| 1. Completed draft <a href="#">report</a>           |                          |
| What do you aim to complete before the next meeting |                          |
| 2. Complete all the remaining UI designs            |                          |
| Supervisor comments                                 |                          |
| 3. Remove commented codes.                          |                          |
| 4. Update UI                                        |                          |

We confirm that the information given in this form is true, complete and accurate.

Student Signature:

Date: 20<sup>th</sup> March, 2023

Supervisor Signature:

Date: 20<sup>th</sup> March, 2023

Figure 9: Logsheet 9

| PROJECT MANAGEMENT LOG                                                                                         |                          |
|----------------------------------------------------------------------------------------------------------------|--------------------------|
| First Name: Ashim                                                                                              | Surname: Bhochhibhoya    |
| Student Number: 2059144                                                                                        | Supervisor: Anil Poudyal |
| Project Title: Khanepa                                                                                         | Month: March             |
| What have you done since the last meeting                                                                      |                          |
| 4. Improved UI for admin dashboard and profile.<br>5. Removed Dead Code.<br>6. Work on professionalism report. |                          |
| What do you aim to complete before the next meeting                                                            |                          |
| 2. Complete the professionalism report.                                                                        |                          |
| Supervisor comments                                                                                            |                          |
| 2. Review <u>on</u> the mentioned parts from the professionalism report.                                       |                          |

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 29<sup>th</sup> March, 2023

Supervisor Signature: 

Date: 29<sup>th</sup> March, 2023

Figure 10: Logsheets 10

Faculty of Science and Engineering  
School of Mathematics and Computer Science



| PROJECT MANAGEMENT LOG                                                                                   |                          |
|----------------------------------------------------------------------------------------------------------|--------------------------|
| First Name: Ashim                                                                                        | Surname: Bhochhibhoya    |
| Student Number: 2059144                                                                                  | Supervisor: Anil Poudyal |
| Project Title: Khanepa                                                                                   | Month: April             |
| What have you done since the last meeting                                                                |                          |
| 1. Completed the professionalism report.<br>2. Completed most of the UI.<br>3. Back end managed properly |                          |
| What do you aim to complete before the next meeting                                                      |                          |
| 1. Make the app function smoothly.<br>2. Work on the backend.                                            |                          |
| Supervisor comments                                                                                      |                          |
| 1. Use Google Map Location's URL instead of API as I had problems integrating API                        |                          |

We confirm that the information given in this form is true, complete, and accurate.

Student Signature:

Date: 6<sup>th</sup> April 2023

Supervisor Signature:

Date: 6<sup>th</sup> April 2023

Figure 11: Logsheet 11

Faculty of Science and Engineering  
School of Mathematics and Computer Science



| PROJECT MANAGEMENT LOG                                                 |                          |
|------------------------------------------------------------------------|--------------------------|
| First Name: Ashim                                                      | Surname: Bhochhibhoya    |
| Student Number: 2059144                                                | Supervisor: Anil Poudyal |
| Project Title: Khanepa                                                 | Month: May               |
| What have you done since the last meeting                              |                          |
| 1. Completed the final report.<br>2. Added sorting for top restaurants |                          |
| What do you aim to complete before the next meeting                    |                          |
| 1. Complete the whole App.<br>2. Start working on poster presentation  |                          |
| Supervisor comments                                                    |                          |
| 1. Review on the progress was done                                     |                          |

We confirm that the information given in this form is true, complete, and accurate.

Student Signature: 

Date: 19<sup>th</sup> April 2023

Supervisor Signature: 

Date: 19<sup>th</sup> April 2023

Figure 12: Logsheets 12

## 7.2 Final Gantt Chart

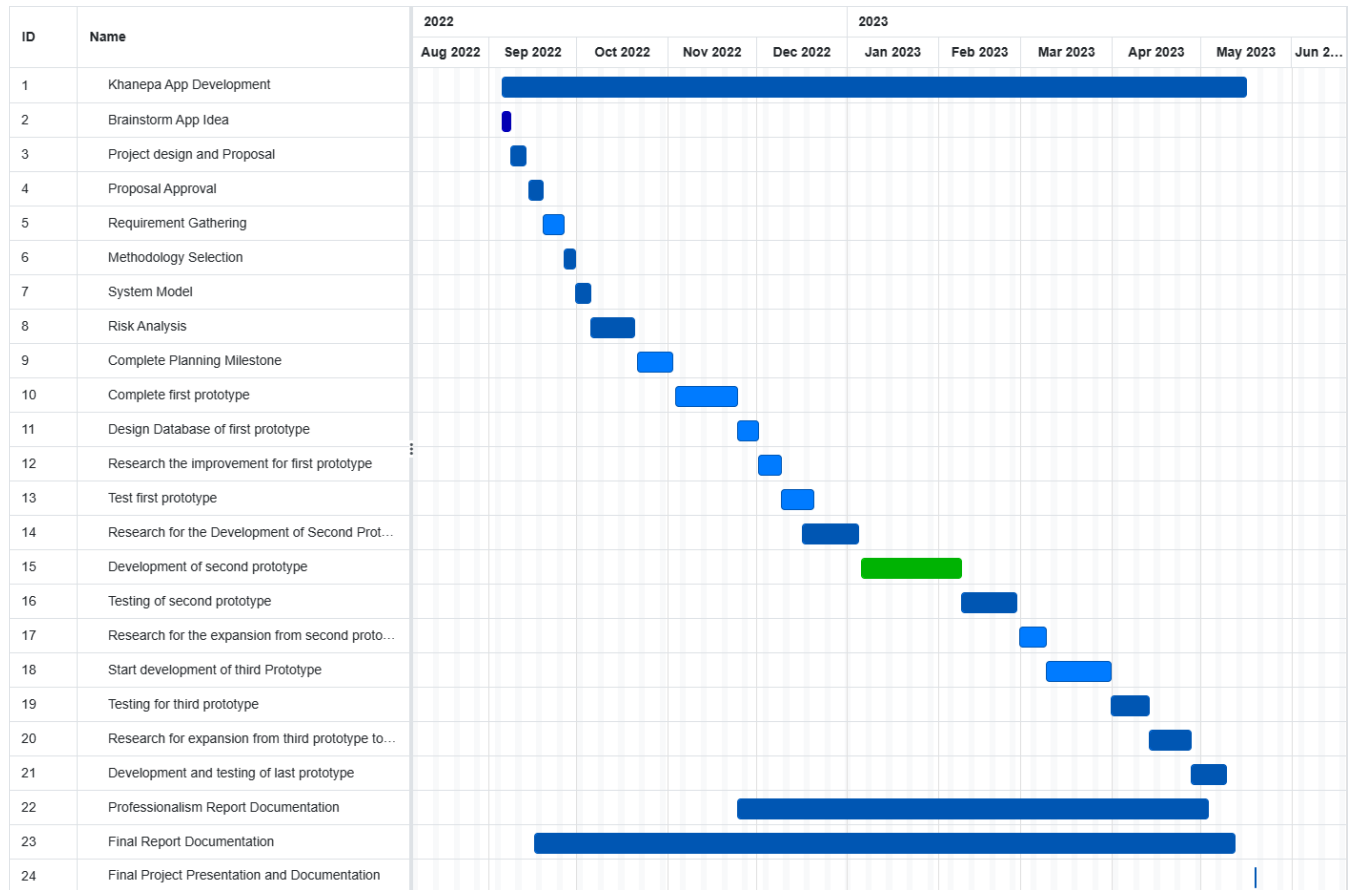


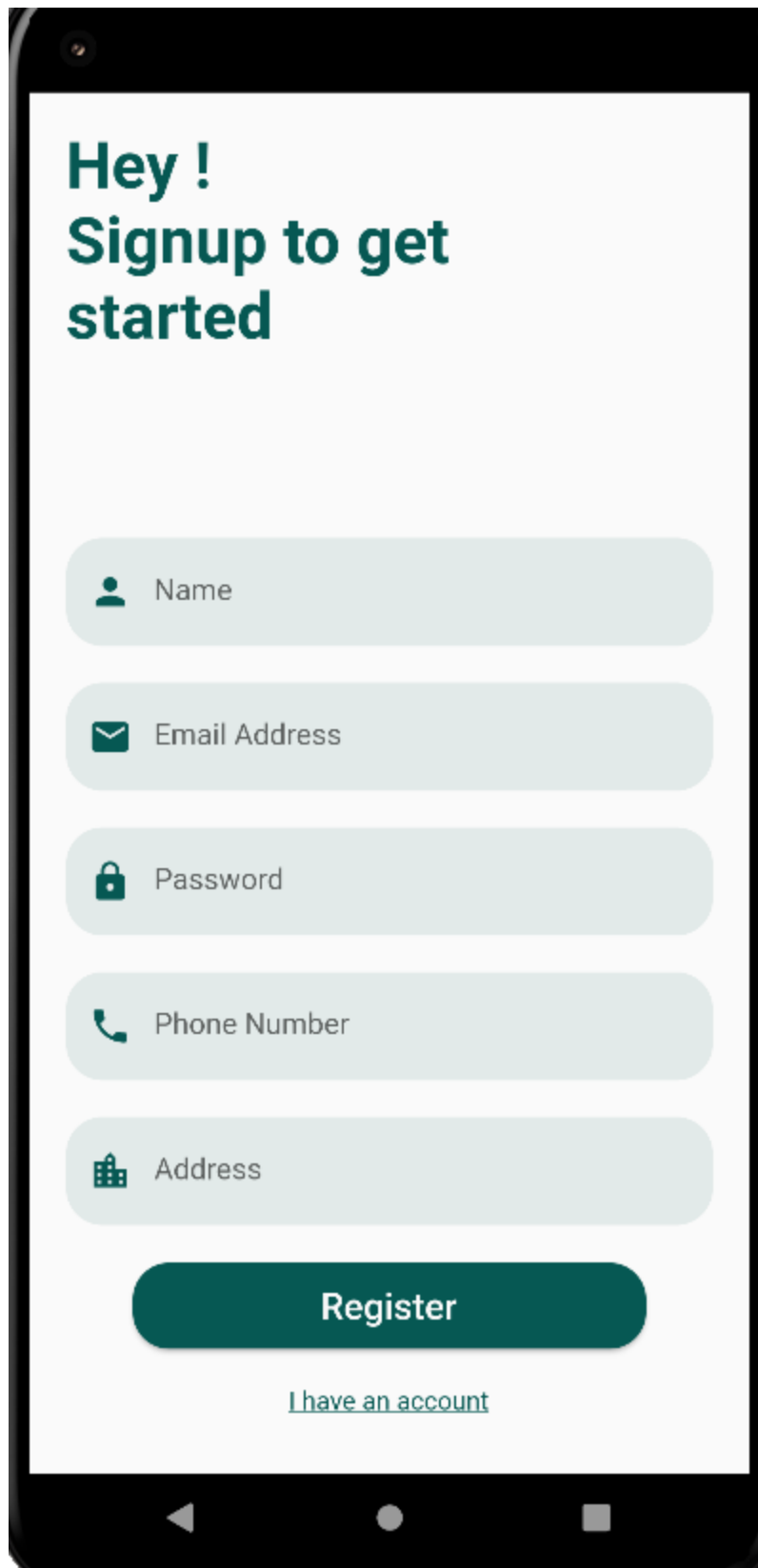
Figure 13: Final Gantt Chart

## 8. Appendices

|                                                                                                 |                                                                                            |                                                                                                        |
|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
|  khanepa-3eee0 |  feedback |  RQ2yAIjSoM8A4vtVBXD4 |
| <a href="#">+ Start collection</a>                                                              | <a href="#">+ Add document</a>                                                             | <a href="#">+ Start collection</a>                                                                     |
| feedback >                                                                                      | Iy5o3Dyggj0AaItWpK0Jn                                                                      | <a href="#">+ Add field</a>                                                                            |
| menu                                                                                            | RQ2yAIjSoM8A4vtVBXD4 >                                                                     | email: "abhochhibhoya@gmail.com"                                                                       |
| order                                                                                           | eaM5U6qkIzYdKrYQCX0K                                                                       | feedback: "good"                                                                                       |
| restaurants                                                                                     |                                                                                            | name: "Ashim"                                                                                          |
| superusers                                                                                      |                                                                                            | userId: "IFKWkPF6ixMkUXJzKelC"                                                                         |
| users                                                                                           |                                                                                            |                                                                                                        |

Figure 14: Firestore Database





The image shows a mobile app's signup screen. At the top, there's a greeting 'Hey !' followed by the main heading 'Signup to get started'. Below this, there are five input fields, each with a small icon on the left: a person icon for 'Name', an envelope icon for 'Email Address', a lock icon for 'Password', a telephone icon for 'Phone Number', and a building icon for 'Address'. Each field is a light gray rounded rectangle. At the bottom of the form is a large, dark green rounded button with the text 'Register' in white. Below the button is a link that says 'I have an account' in a smaller, teal font. The entire screen is framed by a black border, and at the very bottom, there are three small gray icons: a back arrow, a circle, and a square.

Hey !  
Signup to get started

Name

Email Address

Password

Phone Number

Address

Register

[I have an account](#)

Figure 15: Signup Screen of the App

The screenshot shows a mobile application interface for adding a restaurant. At the top, there is a status bar with the time 6:23 and various icons. Below the status bar is a header bar with a back arrow and the title "Add Restaurant". The main content area is titled "Restaurant Information" and contains several input fields: "Restaurant Name" (with a blue border and a cursor), "Description", "Address", "Phone Number", "Email Address", "Password", and "Business Hours". The bottom of the screen shows the Android navigation bar with back, home, and recent apps buttons.

6:23

← Add Restaurant

### Restaurant Information

Restaurant Name

Description

Address

Phone Number

Email Address

Password

Business Hours

Figure 16: Add Restaurant by Admin

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